№493 $\lim\limits_{x\to\frac{\pi}{6}} \frac{2\sin^2x+\sin x-1}{2\sin^2x-3\sin x+1}=\left|\begin{matrix}\sin x=t\\ t\to\frac{1}{2}\end{matrix}\right|=$

$=\lim\limits_{t\to\frac{1}{2}} \frac{2t^2+t-1}{2t^2-3t+1}=\lim\limits_{t\to\frac{1}{2}} \frac{\cancel{(t-\frac{1}{2})}(t+1)}{\cancel{(t-\frac{1}{2})}(t-1)}=-3$

№494 $\lim\limits_{x\to 0} \frac{1-\cos x\cos 2x\cos 3x-\cos x+\cos x}{1-\cos x}=$

$=\lim\limits_{x\to 0} \frac{(1-\cos x)+\cos x(1-\cos 2x\cos 3x)}{1-\cos x}=$

$=\lim\limits_{x\to 0} \left(1+\frac{\cos x}{1-\cos x}\left((1-\cos 2x)+\cos 2x(1-\cos 3x)\right)\right)=$

$=\lim\limits_{x\to 0} \left(1+\frac{2}{x^2}\left(\frac{4x^2}{2}+\frac{9x^2}{2}\right)\right)=\cancel{\lim}\ 1+4+9=14$

№498 $\lim\limits_{x\to\frac{\pi}{4}} \frac{1-ctg^3x}{2-ctg x-ctg^3x}=\left|\begin{matrix}ctg x=t\\ t\to 1\end{matrix}\right|=\lim\limits_{t\to 1} \frac{1-t^3}{2-t-t^3}=$

$=\lim\limits_{t\to 1} \frac{\cancel{(1-t)}(1+t+t^2)}{\cancel{(1-t)}(2+t+t^2)}=\frac{3}{4}$

№499 $\lim\limits_{x\to 0} \frac{\sqrt{1+tg x}-\sqrt{1+\sin x}}{x^3}=\lim\limits_{x\to 0} \frac{1+tg x-1-\sin x}{x^3(\sqrt{1+tg x}-\sqrt{1+\sin x})}=$

$=\lim\limits_{x\to 0} \frac{\sin x(\frac{1}{\cos x}-1)}{x^3\cdot 2}=\lim\limits_{x\to 0} \frac{\cos x-1}{2\cos x\cdot x^2}=\frac{-\frac{x^2}{2}}{2\cos x\cdot x^2}=-\frac{1}{4}$

№402 $\lim \frac{\sqrt{1-\cos x^2}}{1-\cos x} = \frac{\sqrt{\frac{x^4}{2}}}{\frac{x^2}{2}} = \sqrt{2}$

№504 $\lim\limits_{x\to 0} \frac{1-\cos x\sqrt{\cos 2x}\sqrt[3]{\cos 3x}+\cos x-\cos x}{x^2} =$

$+\sqrt{\cos 2x}-\sqrt{\cos 2x}$

$= \lim\limits_{x\to 0} \frac{1-\cos x+\cos x(1-\sqrt{\cos 2x}\sqrt[3]{\cos 3x})}{x^2} =$

$= \lim\limits_{x\to 0} \left(\frac{1-\cos x}{x^2} + \frac{\cos x}{x^2}\left(1-\sqrt{\cos 2x}+\sqrt{\cos 2x}\left(1-\sqrt[3]{\cos 3x}\right)\right)\right) =$

$= \lim\limits_{x\to 0} \left(\frac{1-\cos x}{x^2} - \frac{\cos x(1-\sqrt{\cos 2x})(1+\sqrt{\cos 2x})}{x^2(1+\sqrt{\cos 2x})} + \right.$

$\left. + \frac{\cos x\sqrt{\cos 2x}(1-\sqrt[3]{\cos 3x})(1+\sqrt[3]{\cos 3x}+\sqrt[3]{\cos^2 3x})}{x^2(1+\sqrt[3]{\cos 3x}+\sqrt[3]{\cos^2 3x})}\right) =$

$= \lim\limits_{x\to 0} \left(\frac{1-\cos x}{x^2} - \frac{\cos x(1-\cos 2x)}{x^2 \cdot 2} + \frac{\cos x\sqrt{\cos 2x}(1-\cos 3x)}{3x^2}\right) =$

$= \lim\limits_{x\to 0} \left(\frac{x^2}{2x^2} - \frac{2x^2}{2x^2} + \frac{9x^2}{2\cdot 3x^2}\right) = \frac{1}{2} - 1 + \frac{3}{2} = 3$

№505 $\lim\limits_{x\to +\infty} (\sin\sqrt{x+1} - \sin\sqrt{x}) = \lim\limits_{x\to+\infty} 2\sin\frac{\sqrt{x+1}-\sqrt{x}}{2}\cos\frac{\sqrt{x+1}+\sqrt{x}}{2} =$

$= \lim\limits_{x\to+\infty} 2\sin\underbrace{\overbrace{\frac{1}{2(\sqrt{x+1}+\sqrt{x})}}^{\to 0}}_{\downarrow\, 0}\underbrace{\cos\frac{\sqrt{x+1}+\sqrt{x}}{2}}_{\text{огр}} = 0$

№ 465 № 2 $\lim\limits_{x\to\infty}\left(\frac{x^2+1}{x^2-2}\right)^{x^2}=|1^\infty|=\lim\limits_{x\to\infty}\underbrace{\left(1+\frac{3}{x^2-2}\right)^{\frac{x^2-2}{3}}}_{e}\cdot\frac{x^2}{\frac{x^2-2}{3}}=$

$=e^{\lim\frac{3x^2}{x^2-2}}=e^3$ 471 – 521

№ 471 $\lim\limits_{x\to0}\frac{\sin 5x}{x}=\lim\limits_{x\to0}\frac{5x}{x}=5$

№ 472 $\lim\limits_{x\to\infty}\frac{\sin x}{x}=0$

№ 474 $\lim\limits_{x\to0}\frac{1-\cos x}{x^2}=\frac{\frac{x^2}{2}}{x^2}=\frac{1}{2}$

№ 476 $\lim\limits_{x\to0}\frac{\sin 5x-\sin 3x}{\sin x}=\lim\limits_{x\to0}\frac{2\cancel{\sin x}\cos 4x}{\cancel{\sin x}}=2$

№ 477 $\lim\limits_{x\to0}\frac{\cos x-\cos 3x}{x^2}=\lim\limits_{x\to0}\frac{2\sin 2x\sin x}{x^2}=\lim\limits_{x\to0}\frac{2\cdot2x\cdot x}{x^2}=4$

№ 480 $\lim\limits_{x\to1}(1-x)\operatorname{tg}\frac{\pi x}{2}=\left|\begin{matrix}x-1=t\\t\to0\end{matrix}\right|=\lim\limits_{t\to0}t\left(-\operatorname{tg}\left(\frac{\pi}{2}+\frac{\pi t}{2}\right)\right)=$

$=\lim\limits_{t\to0}t\operatorname{ctg}\frac{\pi t}{2}=\lim\limits_{t\to0}\frac{t}{\operatorname{tg}\frac{\pi t}{2}}=\frac{2\cancel{t}}{\pi\cancel{t}}=\frac{2}{\pi}$

№ 482 $\lim\limits_{x\to a}\frac{\sin x-\sin a}{x-a}=\lim\limits_{x\to a}\frac{2\sin\frac{x-a}{2}\cdot\cos\frac{x+a}{2}}{x-a}=$

$=\lim\limits_{x\to a}\frac{2\cancel{(x-a)}\cdot\cos\frac{2a}{2}\cdot\frac{1}{2}}{\cancel{x-a}}=\cos a$

№483 $\lim\limits_{x\to a}\dfrac{\cos x-\cos a}{x-a}=\lim\limits_{x\to a}\dfrac{-2\sin\frac{x+a}{2}\cdot\sin\frac{x-a}{2}}{x-a}=$

$=\lim\limits_{x\to a}\dfrac{-2\cdot\frac{1}{2}\cdot\cancel{\frac{1}{2}}(x+a)\cancel{(x-a)}}{\cancel{x-a}}=-\sin a$

№484 $\lim\limits_{x\to a}\dfrac{\operatorname{tg}x-\operatorname{tg}a}{x-a}=\left|\operatorname{tg}x-\operatorname{tg}a=\dfrac{\sin(x-a)}{\cos x\cos a}\right|=$

$=\lim\limits_{x\to a}\dfrac{\sin(x-a)}{(x-a)\cos x\cos a}=\dfrac{1}{\cos^2 a}=\sec^2 a$

№485 $\lim\limits_{x\to a}\dfrac{\operatorname{ctg}x-\operatorname{ctg}a}{x-a}=\left|\operatorname{ctg}x-\operatorname{ctg}a=-\dfrac{\sin(x-a)}{\sin x\sin a}\right|=$

$=\lim\limits_{x\to a}-\dfrac{1}{\sin^2 a}=-\operatorname{cosec}^2 a$

№486 $\lim\limits_{x\to a}\dfrac{\sec x-\sec a}{x-a}=\left|\dfrac{1}{\cos x}-\dfrac{1}{\cos a}=\dfrac{\cos a-\cos x}{\cos a\cos x}\right|=$

$=\lim\limits_{x\to a}\dfrac{-2\sin\frac{x+a}{2}\cdot\sin\frac{a-x}{2}}{(x-a)\cos a\cos x}=\dfrac{\sin a}{\cos^2 a}=\sin a\sec^2 a$

№487 $\lim\limits_{x\to a}\dfrac{\operatorname{cosec}x-\operatorname{cosec}a}{x-a}=\left|\dfrac{1}{\sin x}-\dfrac{1}{\sin a}=\dfrac{\sin a-\sin x}{\sin x\sin a}\right|=$

$=\lim\limits_{x\to a}\dfrac{2\sin\cancel{\frac{a-x}{2}}\,\cancel{\sin}\cos\frac{a+x}{2}}{\cancel{(x-a)}\sin x\sin a}=-\dfrac{2\cos a}{\sin^2 a}=-2\cos a\operatorname{cosec}^2 a$

№489 $\lim\limits_{x\to 0}\dfrac{\cos(a+2x)-2\cos(a+x)+\cos a}{x^2}=$

$=\left|(\cos(a+2x)-\cos(a+x))+(\cos a-\cos(a+x))\right|=$

$=-2\sin\dfrac{2a+3x}{2}\cdot\sin\dfrac{x}{2}+(+2\sin\dfrac{2a+x}{2}\cdot\sin(+\dfrac{x}{2}))|=$

$=2\sin\dfrac{x}{2}\left(\sin\dfrac{2a+x}{2}-\sin\dfrac{2a+3x}{2}\right)=2\sin\dfrac{x}{2}\cdot 2\sin(-\dfrac{x}{2})\cdot\cos(a+x)$

$=\lim\limits_{x\to 0}\dfrac{-4\sin^2(\frac{x}{2})\cdot\cos(2x+2a)}{x^2}=-\cos 2a$

№490 $\lim\limits_{x\to 0}\dfrac{tg(a+2x)-2tg(a+x)+tg a}{x^2}=$

$\left|tg\, a+tg B=\dfrac{\sin(a+b)}{\cos a\cdot\cos B}\right|=\lim\limits_{x\to 0}\dfrac{\sin x(\cos a-\cos(a+2x))}{x^2\cos a\cdot\cos(a+2x)\cdot\cos(a+x)}=$

$=\lim\limits_{x\to 0}\dfrac{-2\sin\frac{2a+2x}{2}\cdot\sin(-\frac{x^2}{2})}{x\cos a\cos(x+a)\cos(a+2x)}=\dfrac{2\sin a}{\cos^3 a}$

№491 $\lim\limits_{x\to 0}\dfrac{ctg(a+2x)-2ctg(a+x)+ctg a}{x^2}=$

$=\lim\limits_{x\to 0}\dfrac{(ctg(a+2x)-ctg(a+x))+(ctg a-ctg(a+x))}{x^2}=$

$=\lim\limits_{x\to 0}\dfrac{\sin x(\sin(a+2x)-\sin a)}{x^2\sin a\cdot\sin(a+x)\cdot\sin(a+2x)}=\lim\limits_{x\to 0}\dfrac{2\sin x\cos(a+x)}{x\sin a\cdots}=$

$=\dfrac{2\cos a}{\sin^3 a}$

№492 $$\lim_{x\to0}\frac{\sin(a+x)\sin(a-2x)-\sin^2 a}{x}=$$

$$=\left|\frac{\cos(-x)-\cos(2a+3x)}{2}-\frac{1-\cos 2a}{2}\right|=$$

$$=\lim_{x\to0}\frac{\cos(-x)-\cos(2a+3x)-1+\cos 2a}{2x}=$$

$$=\lim_{x\to0}\frac{-(1-\cos x)-2\sin\frac{4a+3x}{2}\cdot\sin\left(\frac{-3x}{2}\right)}{2x}=$$

$$=\lim_{x\to0}\frac{-\sin^2\frac{x}{2}+2\sin\frac{4a+3x}{2}\cdot\sin\frac{3x}{2}}{2x}=$$

$$=\lim_{x\to0}\left(-\frac{\cancel{\sin}\,x^2}{4\cdot 2x}+\frac{3x}{2x}\cdot\sin\frac{4a+3x}{2}\right)=\frac{3}{2}\sin 2a$$

№493 $$\cancel{\lim_{x\to\frac{\pi}{6}}\frac{2\sin^2x+\sin x-1}{2\sin^2x-3\sin x+1}}$$

№495 $$\lim_{x\to\frac{\pi}{3}}\frac{\sin\left(x-\frac{\pi}{3}\right)}{1-2\cos x}=\left|\begin{matrix}x-\frac{\pi}{3}=t\\ x=t+\frac{\pi}{3}\\ t\to0\end{matrix}\right|=\lim_{t\to0}\frac{\sin t}{1-2\cos\left(t+\frac{\pi}{3}\right)}=$$

$$=\lim_{t\to0}\frac{\sin t}{1-2\left(\cos t\cos\frac{\pi}{3}-\sin t\sin\frac{\pi}{3}\right)}=\lim_{t\to0}\frac{\sin t}{1-\cos t+\sqrt{3}\sin t}=$$

$$=\lim_{t\to0}\frac{\frac{\cancel{t}\sin t}{t}}{\sin\frac{t}{2}\cdot\frac{t}{2}\cdot\frac{\sin\frac{t}{2}}{\frac{t}{2}}+\frac{\cancel{t}\sqrt{3}\sin t}{\cancel{t}}}=\frac{1}{0+\sqrt{3}}=\frac{1}{\sqrt{3}}$$

№496 $\lim\limits_{x\to\frac{\pi}{3}} \frac{tg^3x-3tgx}{\cos(x+\frac{\pi}{6})} = \lim\limits_{x\to\frac{\pi}{3}} \frac{tgx(tg^2x-3)}{\cos x\cdot\cos\frac{\pi}{6}-\sin x\cdot\sin\frac{\pi}{6}} =$

$= \lim\limits_{x\to\frac{\pi}{3}} \frac{tgx\left(\frac{\sin^2x-3\cos^2x}{\cos^2x}\right)}{\frac{\sqrt3}{2}\cos x-\frac{1}{2}\sin x} = \lim\limits_{x\to\frac{\pi}{3}} \frac{tgx(\sin x-\sqrt3\cos x)(\sin x+\sqrt3\cos x)}{-\frac{1}{2}\cos^2x(\sin x-\sqrt3\cos x)}$

$= \lim\limits_{x\to\frac{\pi}{3}} \frac{tgx(\sin x+\sqrt3\cos x)}{-\frac{1}{2}\cos^2x} = \frac{\sqrt3\left(\frac{\sqrt3}{2}+\frac{\sqrt3}{2}\right)}{-\frac{1}{2}\cdot\frac{1}{4}} = -24$

№497 $\lim\limits_{x\to0} \frac{tg(a+x)tg(a-x)-tg^2a}{x^2} = \lim$

$= \lim\limits_{x\to0} \frac{\frac{(tga+tgx)(tga-tgx)}{(1-tga\,tgx)(1+tga\,tgx)}-tg^2a}{x^2} =$

$= \lim\limits_{x\to0} \frac{\frac{tg^2a-tg^2x}{(1-tga\,tgx)^2}-tg^2a}{x^2} = \lim\limits_{x\to0} \frac{\cancel{tg^2a}-tg^2x-\cancel{tg^2a}+tg^4a\,tg^2x}{(1-tg^2x\,tg^2a)\,x^2} =$

$= \lim\limits_{x\to0}\left(\frac{tg^2x}{x^2}\cdot\frac{-1+tg^4a}{1-tg^2a\,tg^2x}\right) = 1\cdot(tg^4a-1)$

№500 $\lim\limits_{x\to0} \frac{x^2}{\sqrt{1+x\sin x}-\sqrt{\cos x}} = \lim\limits_{x\to0} \frac{x^2(\sqrt{1+x\sin x}+\sqrt{\cos x})}{1+x\sin x-\cos x} =$

$= \lim\limits_{x\to0} \frac{x^2(\quad)}{\frac{x^2\sin x}{x}+\frac{1-\cos x}{\frac{x^2}{2}}\cdot\frac{x^2}{2}} = \lim\limits_{x\to0} \frac{\sqrt{1+x\sin x}+\sqrt{\cos x}}{\frac{\sin x}{x}+\frac{\sin^2\frac{x}{2}}{\frac{x^2}{2}}\cdot\frac{1}{2}} =$

$= \frac{1+1}{1+1\cdot\frac{1}{2}} = \frac{2}{\frac{3}{2}} = \frac{4}{3}$

№501 $\lim\limits_{x\to 0}\frac{\sqrt{\cos x}-\sqrt[3]{\cos x}}{\sin^2 x}=\lim\limits_{x\to 0}\frac{\sqrt{\cos x}-\sqrt[3]{\cos x}}{1-\cos^2 x}=$

$=\left|\begin{matrix}\cos x=t\\ t\to 1\end{matrix}\right|=\lim\limits_{t\to 1}\frac{\sqrt{t}-\sqrt[3]{t}}{1-t^2}=\lim\limits_{t\to 1}\frac{t^{\frac{3}{6}}-t^{\frac{2}{6}}}{1-t^2}=$

$=\lim\limits_{t\to 1}\frac{(t^3-t^2)}{(1-t^2)(t^{\frac{5}{3}}+t^{\frac{3}{2}}t^{\frac{1}{3}}+t^{\frac{1}{3}}t^{\frac{2}{3}}+t^{\frac{2}{2}}\cdot t^{\frac{3}{3}}+t^{\frac{1}{2}}t^{\frac{4}{3}}+t^{\frac{5}{3}})}=$

$=\lim\limits_{t\to 1}\frac{-t^2\cancel{(1-t)}}{\cancel{(1-t)}(1+t)(t^{\frac{5}{3}}+\ldots)}=\frac{-1}{2\cdot\underbrace{(1+1)}_{6}}=-\frac{1}{12}$

№503 $\lim\limits_{x\to 0}\frac{1-\sqrt{\cos x}}{1-\cos(\sqrt{x})}=\lim\limits_{x\to 0}\frac{1-\cos x}{(1-\cos\sqrt{x})(1+\sqrt{\cos x})}=$

$=\lim\limits_{x\to 0}\frac{\overbrace{\frac{1-\cos x}{\frac{x^2}{2}}}^{=1}\cdot\frac{x^2}{2}}{\underbrace{\left(\frac{1-\cos\sqrt{x}}{\frac{\sqrt{x}^2}{2}}\right)}_{1}\cdot\frac{\sqrt{x}^2}{2}(1+\sqrt{\cos x})}=\lim\limits_{x\to 0}\frac{x\cdot x}{x\cdot x\cdot(1+\sqrt{\cos x})}=0$

№506 а) $\lim\limits_{x\to 0}\left(\frac{1+x}{2+x}\right)^{\frac{1-\sqrt{x}}{1+x}}=\cancel{\lim(x+}\frac{1}{2}$

б) $\lim\limits_{x\to 1}\left(\frac{1+x}{2+x}\right)^{\frac{1-\sqrt{x}}{1+x}}=\frac{2}{3}^{\lim\limits_{x\to 1}\frac{1-\sqrt{x}}{(1-\sqrt{x})(1+\sqrt{x})}}=\left(\frac{2}{3}\right)^{\frac{1}{2}}=\sqrt{\frac{2}{3}}$

в) $\lim\limits_{x\to +\infty}\left(\frac{1+x}{2+x}\right)^{\frac{1-\sqrt{x}}{1+x}}=\lim\limits_{x\to +\infty}\left(\frac{1+x}{2+x}\right)^{\frac{1}{1+\sqrt{x}}}=1^0=1$

№507 $\lim\limits_{x\to +\infty}\left(\frac{x+2}{2x-1}\right)^{x^2}=0$ №508 $\lim\limits_{x\to\infty}\left(\frac{3x^2-x+1}{2x^2+x+1}\right)^{\frac{x^3}{1-x}}=\infty$

№509 $\lim\limits_{n\to\infty}\left(\sin^n\frac{2\pi n}{3n+1}\right)=0$ №510 $\lim\limits_{x\to\frac{\pi}{4}+0}\left[\operatorname{tg}\left(\frac{\pi}{8}+x\right)\right]^{\operatorname{tg}2x}=0$

№512. $\lim\limits_{x\to\infty}\left(1+\frac{x^2+3}{x^2-2}\right)^{x^2} = e$

№513) $\lim\limits_{x\to 0}(1-2x)^{\frac{1}{x}} = \lim\limits_{x\to 0}(1+(-2x))^{\frac{1}{(-2x)}\cdot(-2x)\cdot\frac{1}{x}} = e^{-2}$

№515 $\lim\limits_{x\to\infty}\left(\frac{x+a}{x-a}\right)^{x} = \lim\limits_{x\to\infty}\left(1+\frac{1}{\frac{x-a}{2a}}\right)^{\frac{x-a}{2a}\cdot 2a+x} = e^{2a}$?

№517 $\lim\limits_{x\to 0}(1+x^2)^{\operatorname{ctg}^2 x} = \lim\limits_{x\to 0}\underbrace{(1+x^2)^{\frac{1}{x^2}}}_{e}{}^{\cdot\frac{x^2}{\sin^2 x}\cos^2 x} = e$

№519 $\lim\limits_{x\to 0}\left(\frac{1+\operatorname{tg} x}{1+\sin x}\right)^{\frac{1}{\sin x}} = \lim\limits_{x\to 0}\left(1+\frac{1}{\frac{1+\sin x}{\operatorname{tg}x-\sin x}}\right)^{\frac{1+\sin x}{\operatorname{tg}x-\sin x}\cdot\frac{1-\cos x}{\cos x(1+\sin x)}} = e^{0}=1$

№520 $\lim\limits_{x\to a}\left(\frac{\sin x}{\sin a}\right)^{\frac{1}{x-a}} = \lim\limits_{x\to a}\left(1+\frac{1}{\frac{\sin a}{\sin x-\sin a}}\right)^{\frac{\sin a}{\sin x-\sin a}\cdot\frac{\sin x-\sin a}{\sin a(x-a)}} =$

$= e^{\operatorname{ctg} a}$

№521 $\lim\limits_{x\to 0}\left(\frac{\cos x}{\cos 2x}\right)^{\frac{1}{x^2}} = \lim\limits_{x\to 0}\left(1+\frac{1}{\frac{\cos 2x}{\cos x-\cos 2x}}\right)^{\frac{\cos 2x}{\cos x-\cos 2x}\cdot\frac{(\cos x-\cos 2x)}{\cos 2x}\cdot\frac{1}{x^2}} =$

$= e^{\frac{3}{2}}$

$\frac{\cos x-\cos 2x}{x^2\,\cancel{\cos 2x}} = \frac{1-\cos x}{x^2}(1+2\cos x) = \frac{1+2\cos x}{2}\cdot\left(\frac{\sin\frac{x}{2}}{\frac{x}{2}}\right)^2 = \frac{3}{2}$

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№528 $\lim\limits_{n\to\infty}\cos^n\frac{x}{\sqrt n} = \lim\limits_{n\to\infty}\left(1+\left(\cos\frac{x}{\sqrt n}-1\right)\right)^n =$

$= \lim\limits_{n\to\infty}\left(1+\left(\cos\frac{x}{\sqrt n}-1\right)\right)^{\frac{n\cdot\left(\cos\frac{x}{\sqrt n}-1\right)}{\cos\frac{x}{\sqrt n}-1}} = e^{\lim\limits_{n\to\infty}-\frac{nx^2}{2n}} = e^{-\frac{x^2}{2}}$

№ 531 $\lim\limits_{x\to a} \frac{\ln x - \ln a}{x-a}, a>0 = \left| \begin{matrix} x-a=t \\ t\to 0 \end{matrix} \right| = \lim\limits_{t\to 0} \frac{\ln(t+a) - \ln(t-t)}{t} =$

$= \lim\limits_{t\to 0} \ln t \cdot \ln a \cdot \frac{1}{t} \cdot \frac{\cancel{\ln t}}{\cancel{\ln t}} \cdot \frac{1}{\ln a} = \lim\limits_{t\to 0} \frac{\ln t}{t} \quad \ln \frac{t+a}{a} = \frac{1}{a}$

№ 537 $\lim\limits_{h\to 0} \frac{\log(\cancel{x}+h) + \log(x-h) - 2\log x}{h^2} =$

$= \lim\limits_{h\to 0} \frac{\log\overset{x^2-h^2}{(\cancel{x}+h)(x-h)} - \cancel{\ }\log x^2}{h^2} = \lim\limits_{h\to 0} \log\left(\frac{x^2-h^2}{x^2}\right) \cdot \frac{1}{h^2} =$

$= \lim\limits_{h\to 0} \cancel{h} \log\left(\cancel{1} - \frac{h^2}{x^2}\right)^{\frac{1}{h^2}} = \cancel{\log} \lim\limits_{h\to 0} \log\left(1 - \frac{h^2}{x^2}\right)^{-\frac{x^2}{h^2} \cdot \frac{(-1)}{x^2}} =$

$= \log e^{\lim\limits_{h\to 0} -\frac{\cancel{h^2}}{x^2}} = \log e^{-\frac{1}{x^2}} \quad (1+x)^{\alpha} - 1) \sim \alpha x$

№ 540 $\lim\limits_{x\to 0} \left| \ln \frac{(nx + \sqrt{1-n^2x^2})}{\cancel{x} + \sqrt{1-x^2}} \right| = 0 \quad (\ln(1+x)) \underset{x\to 0}{\sim} x$

б) $\lim\limits_{x\to 0} \frac{\ln(nx + \sqrt{1-n^2x^2} + 1 - 1)}{\ln(x + \sqrt{1-x^2} + 1 - 1)} = \lim \frac{\cancel{\ln x \cdot \ln(n + \sqrt{\frac{1}{x^2} - n^2})}}{\cancel{\ln x \cdot \ln(1 +}}$

$= \lim\limits_{x\to 0} \frac{nx + \sqrt{1+n^2x^2} - 1}{x + \sqrt{1-x^2} - 1} = \lim\limits_{x\to 0} \frac{\cancel{n^2x^2} - 2nx + x - \cancel{x} + \cancel{n^2x^2}}{x^2 - 2x + x - \cancel{x} + x^2} =$

$= \lim\limits_{x\to 0} \frac{\cancel{2}n^2x^2 - \cancel{2}nx}{\cancel{2}x^2 - \cancel{2}x} = \lim\limits_{x\to 0} \frac{n\cancel{x}(nx-1)}{\cancel{x}(x-1)} = \frac{-n}{-1} = n$

№543 $\lim\limits_{x\to a}\frac{x^x-a^a}{x-a}\overset{a>0}{=}\left|\begin{matrix}x-a=t\\x=a+t\\t\to 0\end{matrix}\right|=\lim\limits_{t\to 0}\frac{(a+t)^{a+t}-a^a}{t}=$

$=\lim\limits_{t\to 0}\frac{(a+t)^{a+t}-a^a+a^{t+a}-a^{t+a}}{t}=\lim\limits_{t\to 0}\frac{(t+a)^{t+a}-a^{t+a}}{t}+\frac{a^{t+a}-a^a}{t}=$

$=\lim\limits_{t\to 0}\left(\frac{a^{t+a}((1+\frac{t}{a})^{t+a}-1)}{t}+\frac{a^a(a^t-1)}{t}\right)=$

$=\lim\limits_{t\to 0}\frac{a^{t+a}\cdot(t+a)\cdot\frac{\cancel{t}}{a}}{\cancel{t}}+\frac{a^a\cdot\cancel{t}\ln a}{\cancel{t}}=\lim\limits_{t\to 0}\frac{a^{t+a}(t+a)}{a}+a^a\ln a=$

$=a^{a}+a^a\ln a=a^a(\ln a+1)=a^a\ln e a$

№544 б) $\lim\limits_{x\to 0}\left(\frac{1+\sin x\cos\alpha x}{1+\sin x\cos\beta x}\right)^{ctg^3x}=$

$=\lim\limits_{x\to 0}\left(1+\frac{1}{\frac{1+\sin x\cos\beta x}{\sin x(\cos\alpha x-\cos\beta x)}}\right)^{ctg^3x\cdot\frac{(1+\sin x\cos\beta x)\cdot\sin x(\cos\alpha x-\cos\beta x)}{(\sin x(\cos\alpha x-\cos\beta x))(1+\sin x\cos\beta x)}}$

$=\cancel{\lim}\ e^{\lim\limits_{x\to 0}\frac{ctg^3x\sin x(\cos\alpha x-\cos\beta x)}{1+\sin x\cos\beta x}}=$

$=e^{\lim\limits_{x\to 0}\frac{\cos^3x\cdot\cancel{\sin x}(\cos\alpha x-\cos\beta x)}{\sin^{\cancel{3}2}x(1+\sin x\cos\beta x)}}=\frac{\cos^3x(-2\sin\frac{\alpha-\beta}{2}\cdot\sin\frac{\alpha+\beta}{2}}{\sin^2x(1+\sin x\cos x}$

$=e^{\lim\frac{-2\cos^3x(\frac{\alpha-\beta}{2})\cdot\frac{(\alpha+\beta)}{2})}{\sin^2x(\ \ 1}}=e^{\lim\frac{\cancel{x}(\alpha^2-\beta^2)}{(\cancel{\alpha^2-\beta^2})tg^2x\,2}}=$

$=e^{-\frac{(\alpha^2-\beta^2)}{2}}$

№548 $\lim\limits_{x\to a}\dfrac{x^{\alpha}-a^{\alpha}}{x^{\beta}-a^{\beta}}=\cancel{\lim}\left|\begin{matrix}x-a=t\\x=a+t\\t\to 0\end{matrix}\right|=\lim\limits_{t\to 0}\dfrac{(a+t)^{\alpha}-a^{\alpha}}{(a+t)^{\beta}-a^{\beta}}=$

$=\lim\limits_{t\to 0}\dfrac{a^{\alpha}((1+\frac{t}{a})^{\alpha}-1)}{a^{\beta}((1+\frac{t}{a})^{\beta}-1)}=\lim\limits_{t\to 0}a^{\alpha-\beta}\dfrac{\frac{t\alpha}{a}}{\frac{t\beta}{a}}=a^{\alpha-\beta}\cdot\dfrac{\alpha}{\beta}$

№522 $\lim\limits_{x\to\frac{\pi}{4}}(tg\,x)^{tg\,2x}=\lim\limits_{x\to\frac{\pi}{4}}\cancel{(1+tg\,x-1)^{\frac{2tg\,x}{1+tg^2x}}}=$

$=\lim\cancel{(1+tg\,x-1)^{\frac{1}{1+tg\,x}\cdot\frac{2tg\,x}{1-tg\,x}}}=\lim\limits_{x\to\frac{\pi}{4}}(tg\,x)^{\frac{2tg\,x}{1-tg^2x}}=$

$=\left|\begin{matrix}t+1=tg\,x\\t\to 0\end{matrix}\right|=\lim\limits_{t\to 0}(t+1)^{\frac{2(t+1)}{-t^2-2t}}=\lim\limits_{t\to 0}(1+t)^{\frac{1}{t}\cdot\frac{2(t+1)}{-t-2}}=e^{-1}$

№523 $\lim\limits_{x\to\frac{\pi}{2}}(\sin x)^{tg\,x}=\lim\limits_{x\to\frac{\pi}{2}}(\sin^2x)^{\frac{tg\,x}{2}}=\cancel{\lim\limits_{x\to\frac{\pi}{2}}(1+ctg^2x)}$

$=\lim\limits_{x\to\frac{\pi}{2}}(1-\cos^2x)^{\frac{\sin x\cos x}{2\cos^2x}}=\left(\frac{1}{e}\right)^{\frac{\sin 2x}{4}}=\left(\frac{1}{e}\right)^{0}=1$

№524 $\lim\limits_{x\to 0}\left(tg\left(\frac{\pi}{4}-x\right)\right)^{ctg\,x}=\left|tg\left(\frac{\pi}{4}-x\right)=\dfrac{1-tg\,x}{1+tg\,x}\right|=$

$=\lim\limits_{x\to 0}\left(\dfrac{1-tg\,x}{1+tg\,x}\right)^{ctg\,x}=\cancel{\lim\left(1+\dfrac{1}{}\right)}\left|\begin{matrix}tg\,x=t\\t\to 0\end{matrix}\right|=$

$=\lim\limits_{t\to 0}\dfrac{(1-t)^{\frac{1}{t}}}{(1+t)^{\frac{1}{t}}}=\dfrac{e^{-1}}{e}=e^{-2}$

№525 $\lim_{x\to\infty}\left(\sin\frac{1}{x}+\cos\frac{1}{x}\right)^{x}=\lim_{t\to 0}(1+\sin t+\cos t-1)^{\frac{1}{t}\cdot\frac{\sin t+\cos t-1}{\sin t+\cos t-1}}=$

$=\lim e^{\lim_{t\to 0}\frac{\sin t+\cos t-1}{t}}=e^{\lim_{t\to 0}\left(\frac{\cos t-1}{t}+\frac{\sin t}{t}\right)}=e^{\lim_{t\to 0}\frac{\cos^2 t-1}{t(\cos t+1)}+1}=$

$=e^{\lim\frac{-\sin^2 t}{t(\cos t+1)}}\quad\lim -\sin t\cdot\frac{\sin t}{t}\cdot\frac{1}{\cos t+1}=0+1=e^{1}=e$

№526 $\lim_{x\to 0}\sqrt[x]{\cos\sqrt{x}}=\lim_{x\to 0}(\cos\sqrt{x})^{\frac{1}{x}}=\lim_{x\to 0}(1+\cos\sqrt{x}-1)^{\frac{1}{x}\cdot\frac{\cos\sqrt{x}-1}{\cos\sqrt{x}-1}}=$

$=e^{\lim_{x\to 0}\frac{\cos\sqrt{x}-1}{x}}=e^{\lim\frac{-2\sin^2\frac{\sqrt{x}}{2}}{x}}=e^{-2\lim\frac{\left(\frac{\sin\frac{\sqrt{x}}{2}}{\frac{\sqrt{x}}{2}}\right)^2}{4}}=e^{-\frac{1}{2}}$

№527 $\lim_{n\to\infty}\left(\frac{n+x}{n-1}\right)^{n}=\lim_{n\to\infty}\left(1+\frac{x+1}{n-1}\right)^{n}=\lim\left(1+\frac{x+1}{n-1}\right)^{\frac{n-1}{x+1}\cdot\frac{x+1}{n-1}\cdot n}=$

$=\lim e^{\lim_{n\to\infty}\frac{n(x+1)}{n-1}}=e^{x+1}$

№529 $\lim_{x\to 0}\frac{\ln(x+1)}{x}=\lim_{x\to 0}\ln\left((1+x)^{\frac{1}{x}}\right)=\ln e=1$

№530 $\lim_{x\to+\infty}x\left[\ln(x+1)-\ln x\right]=\lim_{x\to\infty}\ln\left(\frac{x+1}{x}\right)^{x}=\ln e=1$

№532 $\lim_{x\to+\infty}\left[\sin\ln(x+1)-\sin\ln x\right]=\lim_{x\to+\infty}2\sin\frac{\ln(x+1)-\ln x}{2}\cos\frac{\ln(x+1)+\ln x}{2}=$

$=\sin\frac{1}{2}\lim\ln\left(1+\frac{1}{x}\right)=0\ \Rightarrow\ \lim_{x\to\infty}[\ ]=0$

№535 $\lim_{x\to\infty}\frac{\ln(2+e^{3x})}{\ln(3+e^{2x})}=\frac{3x+}{2x+}=\frac{3}{2}$

№536 $\frac{3}{2}$

№538 $\lim\limits_{x\to 0}\dfrac{\ln\left(tg\left(\frac{\pi}{4}+\alpha x\right)\right)}{\sin\beta x}=\lim\limits_{x\to 0}\left(1+tg\left(\frac{\pi}{4}+\alpha x\right)-1\right)^{\frac{1}{\sin\beta x}}=$

$$=\lim_{x\to 0}\ln\left(1+\frac{\sin\left(\frac{\pi}{4}+\alpha x\right)-\cos\left(\frac{\pi}{4}+\alpha x\right)}{\cos\left(\frac{\pi}{4}+\alpha x\right)}\right)^{\frac{1}{\sin\beta x}}=$$

$$=\lim_{x\to 0}\ln\left(1+\frac{\sqrt{2}\sin\alpha x}{\cos\left(\frac{\pi}{4}+\alpha x\right)}\right)^{\frac{\cos\left(\frac{\pi}{4}+\alpha x\right)}{\sqrt{2}\sin\alpha x}\cdot\frac{\sqrt{2}\sin\alpha x}{\cos\left(\frac{\pi}{4}+\alpha x\right)}\cdot\frac{1}{\sin\beta x}}=$$

$$\frac{\sqrt{2}}{\cos(1)}\cdot\frac{\sin\alpha x}{\sin\alpha x}\cdot\frac{\beta x}{\sin\beta x}\cdot\frac{\alpha}{\beta}$$

$$=\lim_{x\to 0}\ln e^{\frac{2\alpha}{\beta}}=\frac{2\alpha}{\beta}$$

№539 $\lim\limits_{x\to 0}\dfrac{\ln\cos\alpha x}{\ln\cos\beta x}=\lim\dfrac{(\cos\alpha x-1)}{\cos\beta x-1}\cdot\dfrac{\ln\cos\alpha x^{\frac{1}{\cos\alpha x-1}}}{\ln\cos\beta x^{\frac{1}{\cos\beta x-1}}}=$

$$=\lim_{x\to 0}\frac{(\cos\alpha x-1)}{\cos\beta x-1}\cdot\frac{\overbrace{\ln(1+\cos\alpha x-1)^{\frac{1}{\cos\alpha x-1}}}^{=\ln e}}{\underbrace{\ln(1+\cos\beta x-1)^{\frac{1}{\cos\beta x-1}}}_{=\ln e}}=\lim_{x\to 0}\frac{\cos\alpha x-1}{\cos\beta x-1}=$$

$$=\lim_{x\to 0}\frac{2\sin^2\frac{\alpha x}{2}}{2\sin^2\frac{\beta x}{2}}=\lim_{x\to 0}\frac{\left(\frac{\alpha x}{2}\right)^2\left(\frac{\sin\frac{\alpha x}{2}}{\frac{\alpha x}{2}}\right)^2}{\left(\frac{\beta x}{2}\right)^2\left(\frac{\sin\frac{\beta x}{2}}{\frac{\beta x}{2}}\right)^2}=\frac{\alpha^2}{\beta^2}$$

№541 $\lim\limits_{x\to 0}\dfrac{a^x-1}{x}=\left|\begin{matrix}a^x-1=y\\ y\to 0\end{matrix}\right|=\lim\limits_{y\to 0}\dfrac{y}{\log_a(1+y)}=$

$$=\lim_{y\to 0}\frac{1}{\log_a(1+y)^{\frac{1}{y}}}=\frac{1}{\log_a e}=\ln a$$

№542 $\lim\limits_{x\to a}\dfrac{a^x-x^a}{x-a}=\lim\limits_{x\to a}\dfrac{a^x-a^a}{x-a}-\lim\limits_{x\to a}\dfrac{a^a-x^a}{x-a}=a^a\lim\limits_{x\to a}\dfrac{a^{x-a}-1}{x-a}-$

$$-a^{a}\lim_{x\to a}\frac{\left(\frac{x}{a}\right)^{a}-1}{a\left(\frac{x}{a}-1\right)}=a^{a}\lim_{x\to a}\frac{a^{x-a}-1}{x-a}-a^{a}\lim_{x\to a}\frac{\left(\frac{x}{a}\right)^{a}-1}{a\left(\frac{x}{a}-1\right)}=$$

$$=a^{a}\ln a-a^{a}$$

№544 $$\lim_{x\to 0}(x+e^{x})^{\frac{1}{x}}=\lim_{x\to 0}\left(1+\frac{x}{e^{x}}\right)^{\frac{e^{x}}{x}\cdot e^{\frac{x}{e^{x}}}\cdot\frac{1}{x}}=$$

$$=e^{2^{x}}=$$

№546 $$\lim_{n\to\infty}\operatorname{tg}^{n}\left(\frac{\pi}{4}+\frac{1}{n}\right)=\lim_{n\to\infty}\left(\frac{1+\operatorname{tg}\frac{1}{n}}{1-\operatorname{tg}\frac{1}{n}}\right)^{n}=$$

$$=\lim_{n\to\infty}\left(1+\frac{1}{\frac{1-\operatorname{tg}\frac{1}{n}}{2\operatorname{tg}\frac{1}{n}}}\right)^{\frac{1-\operatorname{tg}\frac{1}{n}}{2\operatorname{tg}\frac{1}{n}}\cdot\frac{2\operatorname{tg}\frac{1}{n}}{1-\operatorname{tg}\frac{1}{n}}\cdot n}=e^{2}$$

№547 $$\lim_{x\to 0}\frac{e^{\alpha x}-e^{\beta x}}{\sin\alpha x-\sin\beta x}=\lim_{x\to 0}\frac{e^{\alpha x}-e^{\beta x}}{\alpha x\cdot\frac{\sin\alpha x}{\alpha x}-\beta x\cdot\frac{\sin\beta x}{\beta x}}=$$

$$=\lim_{x\to 0}\frac{e^{\alpha x}-e^{\beta x}}{x(\alpha-\beta)}=\frac{1}{b-a}\lim_{x\to 0}\frac{e^{\beta x}-e^{\alpha x}}{x}=$$

$$=\frac{1}{b-a}\lim_{x\to 0}b\frac{e^{ax}-1}{ax}-a\frac{e^{ax}-1}{ax}=\frac{1}{b-a}\lim b-a=1$$

№549 $$\lim_{x\to b}\frac{a^{x}-a^{b}}{x-b}=\lim a^{b}\frac{a^{x-b}-1}{x-b}=a^{b}\ln a$$

№550 $$\lim_{h\to 0}\frac{a^{x+h}+a^{x-h}-2a^{x}}{h^{2}}=a^{x}\lim\frac{a^{h}+a^{-h}-2}{h^{2}}=$$

$$= \lim_{h\to 0} \frac{a^x}{a^h}\left(\frac{a^h-1}{h}\right)^2 = a^x \ln^2 a$$

№ 551 $$\lim_{x\to\infty} \frac{(x+a)^{x+a}(x+b)^{x+b}}{(x+a+b)^{2x+a+b}} =$$

$$= \lim_{x\to\infty} \frac{(1+\frac{a}{x})^{x+a}(1+\frac{b}{x})^{x+b}}{(1+\frac{a+b}{x})^{2x+a+b}} = \lim_{x\to\infty} \frac{(1+\frac{a}{x})^{\frac{x}{a}\cdot a+a}(1+\frac{b}{x})^{\frac{x}{b}b+b}}{(1+\frac{a+b}{x})^{\frac{x}{a+b}\cdot 2(a+b)-a-b}} =$$

$$= \frac{e^a \cdot e^b}{e^{2(a+b)}} = e^{-a-b}$$

№ 552 $$\lim_{n\to\infty} n(\sqrt[n]{x}-1) = \frac{x^{\frac{1}{n}}-1}{\frac{1}{n}} = \ln x$$

№ 553 $$\lim_{x\to\infty} n^2(\sqrt[n]{x} - \sqrt[n+1]{x}) = \lim_{x\to\infty} \frac{x^{\frac{1}{n}} - x^{\frac{1}{n+1}}}{\frac{1}{n^2}} =$$

$$= \lim \frac{x^{\frac{1}{n+1}}(x^{\frac{1}{n(n+1)}}-1)}{\frac{1}{n(n+1)}\cancel{+(\frac{1}{n^2}-\frac{1}{n^2+n})}} = \lim_{n\to\infty} \frac{\frac{\sqrt[n]{x}-1}{\frac{1}{n}} - \frac{n^2\frac{\sqrt[n]{x}-1}{\frac{1}{n}}}{}}{n+1} =$$

$$= \lim_{n\to\infty} \frac{n\ln x}{n+1} = \ln x \lim_{n\to\infty} \frac{1}{1+\frac{1}{n}} = \ln x$$

№ 554 $$\lim_{n\to\infty} \left(\frac{a-1+\sqrt[n]{b}}{a}\right)^n = \lim_{n\to\infty}\left(1+\frac{b^{\frac{1}{n}}-1}{a}\right)^{\frac{a}{b^{\frac{1}{n}}-1}\cdot\frac{b^{\frac{1}{n}}-1}{\frac{a}{n}}} =$$

$$= e^{\lim_{n\to\infty} \ln b\cdot a} = e^{a\ln b}$$

№555 $\lim_{n\to\infty}\left(\frac{\sqrt[n]{a}+\sqrt[n]{b}}{2}\right)^n = \lim_{n\to\infty}\left(1+\frac{a^{\frac1n}+b^{\frac1n}}{2}-1\right)^{\frac{1}{\frac{a^{1/n}+b^{1/n}}{2}-1}\cdot\left(\frac{a^{1/n}+b^{1/n}}{2}-1\right)\cdot n}$

$= \lim_{n\to\infty} e^{\lim_{n\to\infty}\left(\frac{a^{1/n}+b^{1/n}}{2}-1\right)n} = e^{\frac12(\ln a+\ln b)} = e^{\ln\sqrt{ab}} = \sqrt{ab}$

№556 $\lim_{x\to0}\left(\frac{a^x+b^x+c^x}{3}\right)^{\frac1x}\ \lim_{x\to0}\left(1+\frac{a^x+b^x+c^x}{3}-1\right)^{\frac{1}{\frac{a^x+b^x+c^x}{3}-1}\cdot\left(\frac{a^x+b^x+c^x}{3}-1\right)\cdot\frac1x}$

$= e^{\lim_{x\to0}\left(\frac{a^x+b^x+c^x}{3}-1\right)\cdot\frac1x} = e^{\frac13\ln abc} = \sqrt[3]{abc}$

№557 $\lim_{x\to0}\left(\frac{a^{x+1}+b^{x+1}+c^{x+1}}{a+b+c}\right)^{\frac1x} =$

$=\lim_{x\to0}\frac{a^{x+1}+b^{x+1}+c^{x+1}-a-b-c}{x(a+b+c)} = \frac{1}{a+b+c}\lim\left(\frac{a\cdot(a^{x+1}-1)}{x}+b\cdot+c\cdot\right)=$

$=\cancel{\frac{1}{a+b+c}}\ \overset{\ln}{\cancel{\lim}}(a^a b^b c^c)\cdot\frac{1}{abc}$

lim $\lim_{x\to0}\left(1+\frac{a^{x+1}+b^{x+1}+c^{x+1}}{a+b+c}-1\right)^{\frac{1}{\frac{a^{x+1}+b^{x+1}+c^{x+1}}{a+b+c}-1}\cdot\left(\frac{a^{x+1}+b^{x+1}+c^{x+1}}{a+b+c}-1\right)\cdot\frac1x}=$

$= e^{\lim_{x\to0}\frac1x\left(\frac{a^{x+1}+b^{x+1}+c^{x+1}}{a+b+c}-1\right)} = e^{\frac{1}{a+b+c}\cdot\lim_{x\to0}\frac{a(a^x-1)}{x}+b\cdot+c\cdot}=$

$= e^{\frac{1}{a+b+c}\cdot\ln(a^a\cdot b^b\cdot c^c)}$

№558 $\lim_{x\to0}\left(\frac{a^{x^2}+b^{x^2}}{a^x+b^x}\right)^{\frac1x} = \lim_{x\to0}\left(1+\frac{1}{\frac{a^x+b^x}{a^{x^2}+b^{x^2}-a^x-b^x}}\right)^{\frac{a^x+b^x}{(\ )}\cdot\frac{(\ )}{a^x+b^x}\cdot\frac1x}=$

$= e^{\lim_{x\to0}\frac{1}{a^x+b^x}\cdot\left(\frac{a^{x^2}-a^x}{x}+\frac{b^{x^2}-b^x}{x}\right)} = e^{\frac12\lim x\frac{b^x-1}{x}-\frac{b^x-1}{x}+x\frac{a^x-1}{x}-\frac{a^x-1}{x}}=$

$= e^{-\frac12\ln ab} = \frac{1}{\sqrt{ab}}$

№559 $\lim\limits_{x\to 0} \frac{a^{x^2}-b^{x^2}}{(a^x-b^x)^2} = \lim \left(\frac{a^{x^2}-1}{x^2} - \frac{b^{x^2}-1}{x^2}\right) \cdot \frac{1}{\left(\frac{a^x-1}{x} - \frac{b^x-1}{x}\right)^2} =$

$= (\ln a - \ln b) \cdot \frac{1}{(\ln a - \ln b)^2} = \frac{1}{\ln a - \ln b} = \left(\ln \frac{a}{b}\right)^{-1}$

№559.2 $\lim\limits_{x\to 1} \frac{\sin \pi (x)^\alpha}{\sin \pi (x)^\beta} = \left| \begin{array}{l} 1-x=t \\ x=1-t \\ t\to 0 \end{array} \right| = \lim\limits_{t\to 0} \frac{\sin \pi((1-t)^\alpha + 1 - 1)}{\sin \pi((1-t)^\beta + 1 - 1)} =$

$= \lim\limits_{t\to 0} \frac{+\sin \pi((1-t)^\alpha - 1)}{+\sin \pi((1-t)^\beta - 1)} = \frac{\pi \alpha t}{\pi \beta t} = \frac{\alpha}{\beta}$

№560 $\lim\limits_{x\to a} \frac{a^{a^x} - a^{x^a}}{a^x - x^a} = \left| \begin{array}{l} x-a=t \\ t\to 0 \end{array} \right| = \lim\limits_{t\to 0} \frac{a^{a^{(t+a)}} - a^{(t+a)^a}}{a^{(t+a)} - (t+a)^a} =$

$= \lim\limits_{t\to 0} \frac{a^{a^a}\left(a^{a^a(a^t-1)} - a^{(t+a)^a - a^a}\right)}{a^a\left(\underbrace{a^t-1}_{\sim t\ln a} - \underbrace{(1+\frac{t}{a})^a - 1}_{\sim t}\right)} = \cancel{A}$

$a^{a^t \cdot a^a} + a^{(t+a)} = a^{a^a}\left(a^{a^a(a^t-1)} + a^{(t+a)^a - a^a}\right) =$

$= a^{a^a}\left(a^{a^a(a^t-1)} + a^{a^a((1+\frac{t}{a})^a - 1)}\right)$

$\cancel{A} = \lim\limits_{t\to 0} \frac{a^{a^a}\left(a^{a^a(a^t-1)} - 1 + 1 - a^{(1+\frac{t}{a})^a - 1}\right)}{a^a \cdot t(\ln a - 1)} =$

$= \frac{a^{a^a}}{a^a} \lim \frac{a^a(a^t-1)\ln a - ((1+\frac{t}{a})^a - 1)\ln a}{t(\ln a - 1)} =$

$$= \frac{a^{a^a}\ln a}{a^a} \lim_{t\to 0} \frac{\ln a\, a^a t - t}{t(\ln a - 1)} = \frac{a^{a^a}\ln a}{a^a}\left(\frac{a^a \ln a - 1}{\ln a - 1}\right)$$

№ 567 $\lim_{x\to 0} \frac{\ln(1+xe^x)}{\ln(x+\sqrt{1+x^2})} = \lim_{x\to 0} \frac{xe^x}{x+\sqrt{1+x^2}-1} =$

$\ln(1+x) \sim x$

$$= \lim_{x\to 0} \frac{xe^x(x+\sqrt{1+x^2}+1)}{(x+\sqrt{1+x^2})^2-1} =$$

$$\lim_{x\to 0} \frac{xe^x(x-1-\sqrt{x^2+1})}{(x+1)^2-1-x^2} = \lim_{x\to 0} \frac{xe^x(x-1-\sqrt{1+x^2})}{x^2+2x+1-1-x^2} = \frac{e^0(-2)}{(-2)} = 1$$

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№ 570 $\lim_{x\to+\infty}\left(\ln\frac{x+\sqrt{x^2+1}}{x+\sqrt{x^2-1}} \cdot \ln^{-2}\frac{x+1}{x-1}\right) = \left(\frac{0}{0}\right) =$

$$= \lim_{x\to+\infty}\left(\frac{\ln\left(1+\frac{\sqrt{x^2+1}-\sqrt{x^2-1}}{x+\sqrt{x^2-1}}\right)}{\ln^2\left(1+\frac{2}{x+1}\right)}\right) = \lim_{x\to\infty} \frac{(\sqrt{x^2+1}-\sqrt{x^2-1})(x+1)^2}{(x+\sqrt{x^2-1})\cdot 4} =$$

$$= \lim_{x\to+\infty} \frac{(x+1)^2(2)}{4(\sqrt{x^2+1}+\sqrt{x^2-1})(x+\sqrt{x^2-1})} = \lim_{x\to+\infty} \frac{x^2(1+\frac{1}{x})^2}{x^2(\sqrt{1+\frac{1}{x^2}}+\sqrt{1-\frac{1}{x^2}})x(1+\sqrt{1-\frac{1}{x^2}})}$$

$$= \frac{1}{8}$$

№577 а) ~~lim~~ ~~sh~~ x^2

№576,1 $\lim\limits_{x\to 0} \frac{sh^2 x}{\ln(ch 3x)} = \lim\limits_{x\to 0} \frac{sh^2 x}{ch 3x - 1} =$

$= \lim\limits_{x\to 0} \frac{sh^2 x}{2sh^2 \frac{3x}{2}} = \frac{x^2 \cdot 4}{2 \cdot 9x^2} = \frac{2}{9}$

№580 $\lim\limits_{n\to\infty} \left(\frac{ch\frac{\pi}{n}}{\cos\frac{\pi}{n}}\right)^{n^2} = \lim\limits_{n\to\infty}\left(1 + \frac{ch\frac{\pi}{n} - \cos\frac{\pi}{n}}{\cos\frac{\pi}{n}}\right)^{n^2} =$

$= e^{\lim\limits_{n\to\infty} \frac{ch\frac{\pi}{n} - \cos\frac{\pi}{n}}{\cos\frac{\pi}{n}} \cdot n^2} = e^{\lim\limits_{n\to\infty} \frac{\left(1+\frac{\pi^2}{2n^2} - \left(1 - \frac{\pi^2}{2n^2}\right)\right)}{1} \cdot n^2} =$

$ch\, x = 1 - \frac{x^2}{2}$ $= e^{\pi^2}$

$= e$

№581

$\lim\limits_{x\to\infty} \arcsin \frac{1-x}{1+x} = \arcsin(-1) = -\frac{\pi}{2}$

№569 $\lim\limits_{x\to+0} \left[\ln(x\ln a) \cdot \ln\left(\frac{\ln(ax)}{\ln\left(\frac{x}{a}\right)}\right)\right]$ $a>1$

$= \lim\limits_{x\to+0} \ln\left(\frac{\ln a + \ln x}{\ln x - \ln a}\right)^{\ln x + \ln\ln a} = \lim\limits_{x\to+0} \ln\left(1 + \frac{2\ln a}{\ln x - \ln a}\right)^{\frac{\ln x \ln a}{2\ln a}} =$

$= \lim \ln e^{\lim\limits_{x\to+0} \frac{2\ln a(\ln x - \ln(\ln a))}{\ln x - \ln a}} = \ln e^{\ln a^2} = \ln a^2$

№571 $\lim\limits_{x\to 0}\frac{\sqrt{1+x\sin x}-1}{e^{x^2}-1}=\lim\limits_{x\to 0}\frac{x\sin x}{(e^{x^2}-1)(1+\sqrt{1+x\sin x})}=$

$=\lim\limits_{x\to 0}\frac{\frac{\sin x}{x}}{\left(\frac{e^{x^2}-1}{x^2}\right)(1+\sqrt{1+x\sin x})}=\frac{1}{2}$

№572 $\lim\limits_{x\to 0}\frac{\cos(xe^{x})-\cos(xe^{-x})}{x^3}=$

$=\lim\limits_{x\to 0}\frac{-2\sin\left(\frac{x(e^{x}+e^{-x})}{2}\right)\cdot\sin\left(\frac{x(e^{x}-e^{-x})}{2}\right)}{x^3}=$

$=-2\lim\limits_{x\to 0}\frac{\sin\frac{x(e^{x}+e^{-x})}{2}}{\frac{x(e^{x}+e^{-x})}{2}}\cdot\frac{\sin\frac{x(e^{x}-e^{-x})}{2}}{\frac{x(e^{x}-e^{-x})}{2}}\cdot\frac{x^2(e^{4x}-1)}{4x^3e^{2x}}=$

$=-2\lim\limits_{x\to 0}\frac{e^{4x}-1}{4x}\cdot\frac{1}{e^{2x}}=-2$

№573 $\lim\limits_{x\to 0}\left(2e^{\frac{x}{x+1}}-1\right)^{\frac{x^2-1}{x}}=\lim\limits_{x\to 0}\left(1-2(e^{\frac{x}{x+1}}-1)\right)^{1\left(\frac{1}{2(e^{\frac{x}{x+1}}-1)}\right)\cdot\frac{2(e^{\frac{x}{x+1}}-1)(x^2-1)}{x}}=e^{\lim\limits_{x\to 0}\frac{2(x-1)(x+1)(e^{\frac{x}{x+1}}-1)}{x}}=$

$=e^{-2\lim\limits_{x\to 0}\frac{e^{\frac{x}{x+1}}-1}{\frac{x}{x+1}}}=e^{-2}$

№574 $\lim\limits_{x\to1}(2-x)^{\sec\frac{\pi x}{2}} = \lim\limits_{x\to1}(1-x+1)^{\frac{1}{1-x}\cdot\frac{1-x}{\cos\frac{\pi x}{2}}} =$

$= \left|\begin{matrix} x-1=t \\ t\to0\end{matrix}\right| = e^{\lim\limits_{t\to0}\frac{-t}{\sin\frac{\pi t}{2}}} = e^{\lim \frac{2}{\frac{\sin\frac{\pi t}{2}}{\frac{\pi t}{2}}\cdot\pi}} = e^{\frac{2}{\pi}}$

№575 $\lim\limits_{x\to\frac{\pi}{2}} \frac{1-\sin^{\alpha+\beta}x}{\sqrt{(1-\sin^{\alpha}x)(1-\sin^{\beta}x)}} =$

$= \lim\limits_{x\to\frac{\pi}{2}} \frac{1-e^{(\alpha+\beta)\ln(\sin x)}}{\sqrt{(1-e^{\alpha\ln\sin x})(1-e^{\beta\ln\sin x})}} =$

$= \lim\limits_{x\to\frac{\pi}{2}} \left(\frac{1-e^{(\alpha+\beta)\ln(\sin x)}}{(\alpha+\beta)\ln\sin x}\right)\cdot\left(\frac{\alpha\ln\sin x}{1-e^{\alpha\ln\sin x}}\right)^{\frac{1}{2}}\cdot$

$\cdot\left(\frac{\beta\ln\sin x}{1-e^{\beta\ln\sin x}}\right)^{\frac{1}{2}}\left(\frac{(\alpha+\beta)\ln\sin x}{\sqrt{\alpha\beta}\,(\ln\sin x)}\right) = \frac{\alpha+\beta}{\sqrt{\alpha\beta}}$

№576 а) $\lim\limits_{x\to0}\frac{\operatorname{sh}x}{x} = 1$

б) $\lim\limits_{x\to0}\frac{\operatorname{ch}x-1}{x^2} = \lim\limits_{x\to0}\frac{\frac{e^x+e^{-x}}{2}-1}{x^2} = \lim\limits_{x\to0}\frac{(e^x-1)^2}{x^2\cdot2e^x} =$

$= \frac{1}{2}\lim\limits_{x\to0}\frac{(e^x-1)^2}{x^2} = \frac{1}{2}$

в) $\lim\limits_{x\to0}\frac{\operatorname{th}x}{x} = 1$

№577 б) $\lim\limits_{x\to+\infty} \frac{\operatorname{sh}\sqrt{x^2+x} - \operatorname{sh}\sqrt{x^2-x}}{\operatorname{ch} x} =$

$= \lim\limits_{x\to+\infty} \frac{2\operatorname{sh}\frac{\sqrt{x^2+x}-\sqrt{x^2-x}}{2}\cdot \operatorname{ch}\frac{\sqrt{x^2+x}+\sqrt{x^2-x}}{2}}{\operatorname{ch} x} =$

$= 2\lim\limits_{x\to+\infty} \frac{\operatorname{sh}\frac{x}{\sqrt{x^2+x}+\sqrt{x^2-x}}\cdot \operatorname{ch}\frac{\sqrt{x^2+x}+\sqrt{x^2-x}}{2}}{\operatorname{ch} x} =$

$= 2\operatorname{sh}\frac{1}{2}\lim\limits_{x\to\infty} \frac{e^{\frac{\sqrt{x^2+x}+\sqrt{x^2-x}}{2}} + e^{-\frac{\sqrt{x^2+x}+\sqrt{x^2-x}}{2}}}{e^x+e^{-x}} = 2\operatorname{sh}\frac{1}{2}$

№578 а) $\lim\limits_{x\to a} \frac{\operatorname{sh}x-\operatorname{sh}a}{x-a} = \lim\limits_{x\to a} \frac{2\operatorname{sh}\frac{x-a}{2}\operatorname{ch}\frac{x+a}{2}}{x-a} = \operatorname{ch} a$

б) $\lim\limits_{x\to a} \frac{\operatorname{ch}x-\operatorname{ch}a}{x-a} = \lim\limits_{x\to a} \frac{-2\operatorname{sh}\frac{x+a}{2}\cdot \operatorname{sh}\frac{x-a}{2}}{x-a} = -\operatorname{sh}a$

в) $\lim\limits_{x\to 0} \frac{\ln\operatorname{ch}x}{\ln\cos x} = \lim\limits_{x\to 0} \frac{\operatorname{ch}x-1}{\cos x-1}\cdot\frac{\ln \operatorname{ch}x^{\frac{1}{\operatorname{ch}x-1}}}{\ln\cos x^{\frac{1}{\cos x-1}}} =$

$= \lim\limits_{x\to 0} \frac{\operatorname{ch}x-1}{\cos x-1}\cdot\frac{\ln(1+\operatorname{ch}x-1)^{\frac{1}{\operatorname{ch}x-1}}}{\ln(1+\cos x-1)^{\frac{1}{\cos x-1}}} = \lim\limits_{x\to 0}\frac{\operatorname{ch}x-1}{-\sin^2\frac{x}{2}} =$

$= \lim\limits_{x\to 0}\frac{1}{2}\frac{4(1-\operatorname{ch}x)}{\frac{(\sin^2\frac{x}{2})^2}{(\frac{x}{2})^2}\cdot x^2} = \lim\limits_{x\to 0}\frac{1}{2}\frac{4(1-\operatorname{ch}x)}{x^2} = \lim\limits_{x\to 0}\frac{4-2(e^x+e^{-x})}{x^2}\frac{1}{2} =$

$= \frac{1}{2}\lim\limits_{x\to 0} -\frac{2(e^x-1)^2}{x^2e^x} = -\lim\limits_{x\to 0}\left(\frac{e^x-1}{x}\right)^2 = -1$

579 а) $\lim\limits_{x\to+\infty}(x-\ln ch x)=\lim\limits_{x\to+\infty}\left(x-\ln\frac{e^x+e^{-x}}{2}\right)=$

$=\lim\limits_{x\to+\infty}\left(x+\ln 2-\ln\left(e^x+\frac{1}{e^x}\right)\right)=\lim\limits_{x\to+\infty}\left(\ln 2+2x+\ln(e^{2x}+1)\right)=$

$=\lim\limits_{x\to+\infty}\left(\ln 2-\frac{\left(\ln\left(1+\frac{1}{e^{2x}}\right)^{e^{2x}}\right)}{e^{2x}}\right)=\lim\limits_{x\to+\infty}\ln 2-\frac{\ln e}{e^{2x}}=$

$=\ln 2$

№582 $\lim\limits_{x\to+\infty}\arccos\left(\sqrt{x^2+x}-x\right)=\lim\limits_{x\to+\infty}\arccos\frac{x}{\sqrt{x^2+x}+x}=$

$=\arccos\frac{1}{2}=\frac{\pi}{3}$

№583 $\lim\limits_{x\to 2}\operatorname{arctg}\frac{x-4}{(x-2)^2}=\pm\frac{\pi}{2}$?

№584 $\lim\limits_{x\to-\infty}\operatorname{arctg}\frac{x}{\sqrt{1+x^2}}=-\frac{\pi}{4}$

№585 $\lim\limits_{h\to 0}\frac{\operatorname{arctg}(x+h)-\operatorname{arctg}x}{h}=$

$=\lim\limits_{h\to 0}\frac{\frac{\operatorname{arctg}h}{1+x(x+h)}}{\frac{h}{1+x(x+h)}}\cdot\frac{1}{1+x(x+h)}=\frac{1}{1+x^2}$

№586 $\lim\limits_{x\to 0}\frac{\ln\frac{1+x}{1-x}}{\operatorname{arctg}(1+x)-\operatorname{arctg}(1-x)}=$

$$= \lim_{x\to 0} \ln\left(1+\frac{2x}{1-x}\right) \cdot \frac{1}{\operatorname{arctg}\frac{2x}{2-x^2}} = \lim_{x\to 0} \frac{\ln\left(1+\frac{2x}{1-x}\right)}{\frac{2x}{1-x}} \cdot \frac{\frac{2x}{2-x^2}}{\operatorname{arctg}\frac{2x}{2-x^2}} \cdot \frac{\frac{2x}{1-x}}{\frac{2x}{2-x^2}}$$

$$= 2$$

№588 $\lim\limits_{x\to+\infty} x\left(\frac{\pi}{2} - \arcsin\frac{x}{\sqrt{x^2+1}}\right) = \lim\limits_{x\to+\infty} x \arcsin\frac{1}{\sqrt{x^2+1}} =$

$$= \lim_{x\to+\infty} \frac{\arcsin\frac{1}{\sqrt{x^2+1}}}{\frac{1}{\sqrt{x^2+1}}} \cdot \frac{x}{\sqrt{x^2+1}} = 1$$

№589 $\lim\limits_{x\to\infty} x\left(\frac{\pi}{4} - \operatorname{arctg}\frac{x}{x+1}\right) = \lim\limits_{x\to\infty} x\left(\operatorname{arctg}1 - \operatorname{arctg}\frac{x}{x+1}\right) =$

$$= \lim_{x\to\infty} x \operatorname{arctg}\frac{\frac{1}{x+1}}{\frac{2x+1}{x+1}} = \lim_{x\to\infty} x \operatorname{arctg}\frac{1}{2x+1} = \lim_{x\to\infty} \frac{\operatorname{arctg}\frac{1}{2x+1}}{\frac{1}{2x+1}} \cdot \frac{x}{2x+1} =$$

$$= \frac{1}{2}$$

№591 а) $\lim\limits_{x\to 0} \frac{1}{x^{100}} e^{-\frac{1}{x^2}} = \left|\begin{matrix}\frac{1}{x}=t \\ t\to\infty\end{matrix}\right| = \lim\limits_{t\to\infty} \frac{t^{100}}{e^{t^2}} = 0$

б) $\lim\limits_{x\to+0} x\ln x = \left|\begin{matrix}t=\frac{1}{x} \\ t\to+\infty\end{matrix}\right| = \lim\limits_{t\to+\infty} \frac{-\ln t}{t} = 0$

№592 а) $\lim\limits_{x\to-\infty} \sqrt{x^2+x} - x = \left|\begin{matrix}-x=t \\ t\to+\infty\end{matrix}\right| = \lim\limits_{t\to+\infty} \sqrt{t^2-t} + t = +\infty$

б) $\lim\limits_{x\to+\infty} \sqrt{x^2+x} - x = \lim\limits_{x\to+\infty} \frac{x}{\sqrt{x^2+x}+x} = \frac{1}{2}$

№593 а) $\lim\limits_{x\to-\infty} \sqrt{1+x+x^2} + \sqrt{1-x-x^2} = \left| \begin{matrix} -x=t \\ t\to+\infty \end{matrix} \right| =$

$= \lim\limits_{t\to+\infty} \frac{-2t}{\sqrt{1-t+t^2} + \sqrt{1+t-t^2}} = -1$

б) $\lim\limits_{x\to+\infty} \sqrt{1+x+x^2} - \sqrt{1-x+x^2} = \lim\limits_{x\to+\infty} \frac{1+x+x^2-1+x-x^2}{\sqrt{1+x+x^2}+\sqrt{1-x+x^2}} =$

$= \lim\limits_{x\to+\infty} \frac{2x}{x\sqrt{1}+x\sqrt{1}} = 1$

№595 а) $\lim\limits_{x\to1-0} \operatorname{arctg} \frac{1}{1-x} = \frac{\pi}{2}$

б) $\lim\limits_{x\to1+0} \operatorname{arctg} \frac{1}{1-x} = -\frac{\pi}{2}$

№596 а) $\lim\limits_{x\to-0} \frac{1}{1+e^{\frac{1}{x}}} = 1$?

б) $\lim\limits_{x\to+0} \frac{1}{1+e^{\frac{1}{x}}} = 0$

№597 а) $\lim\limits_{x\to-\infty} \frac{\ln(1+e^x)}{x} = 0$

?

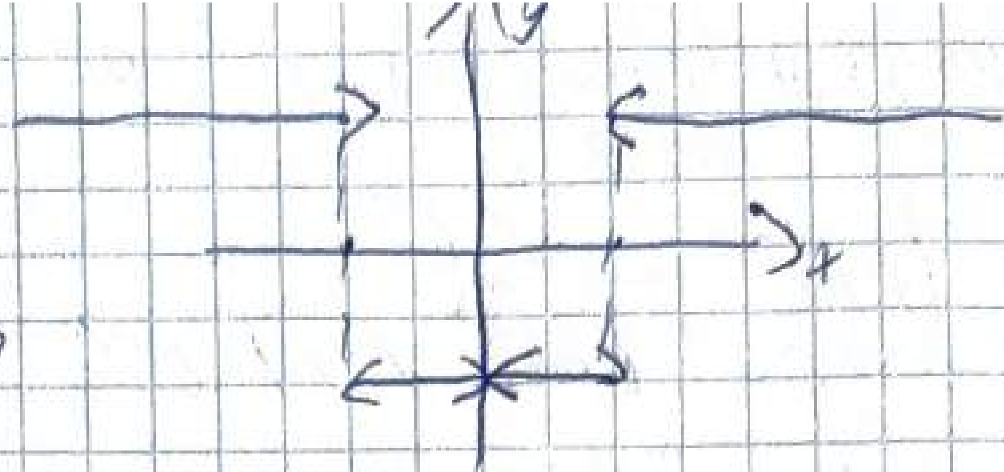
б) $\lim\limits_{x\to+\infty} \frac{\ln(1+e^x)}{x} = 1$

615

$$y=\lim_{n\to\infty}\frac{x^n-x^{-n}}{x^n+x^{-n}}$$

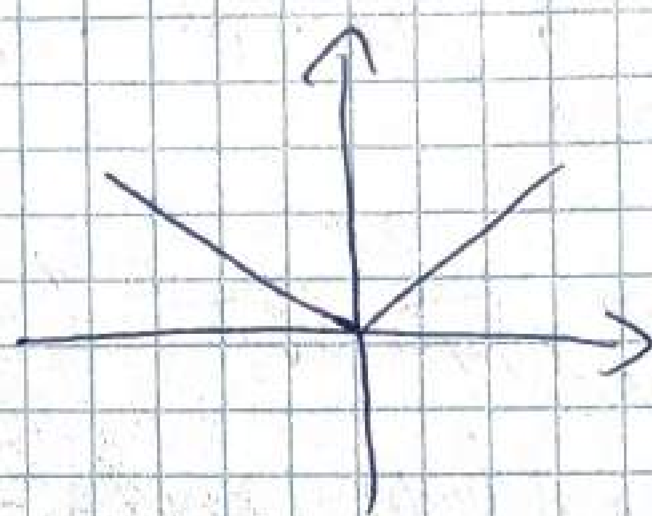
$|x|<1$ $y=-1$ $y(1)=0$

$|x|>1$ $y=1$



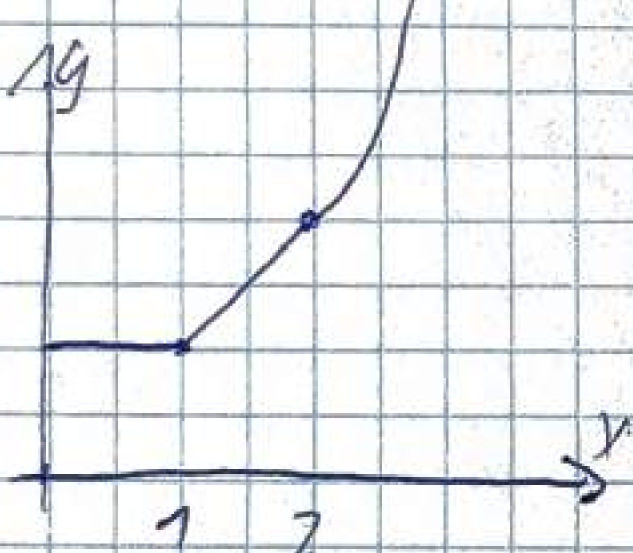
№616

$$\cancel{y}=\lim_{x\to\infty}\sqrt{x^2+\frac{1}{n^2}}$$



№618

$$y=\lim_{n\to\infty}\sqrt[n]{1+x^n+\left(\frac{x^2}{2}\right)^n}$$ $x\geq 0$



$x\in(0,1)\Rightarrow y\to 1$

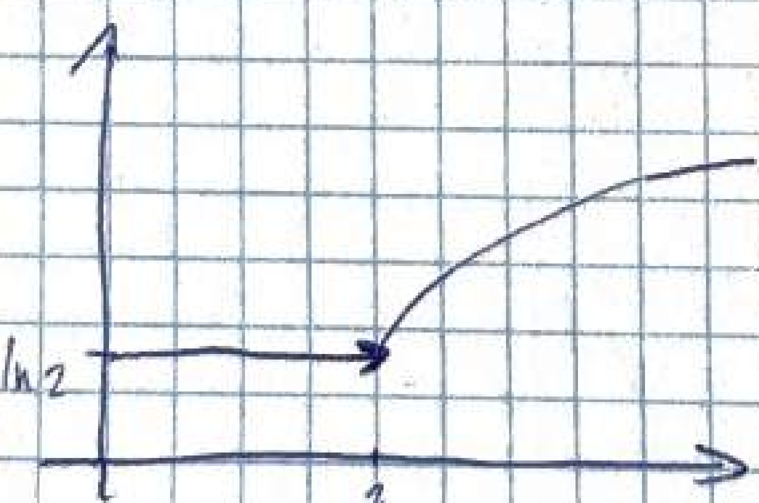
$x\in[1,2)\Rightarrow y=x$

$x\geq 2\Rightarrow y=\frac{x^2}{2}$

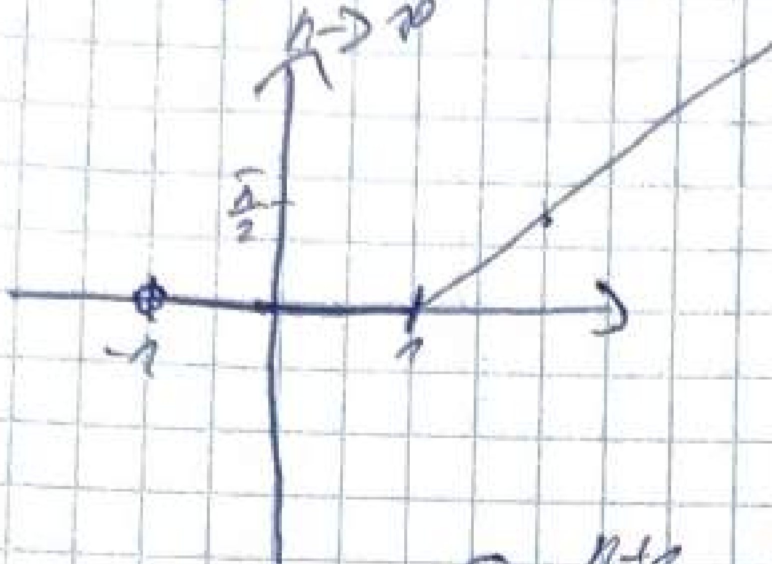
№621

$$y=\lim_{n\to\infty}\frac{\ln(2^n+x^n)}{n}$$ $x\geq 0$

$x\in(0,2]\Rightarrow y=\ln 2$



№622 $y=\lim\limits_{n\to\infty}(x-1)\,\mathrm{arctg}\,x^n$



$|x|<1$

$x>1 \Rightarrow y=\frac{\pi}{2}(x-1)$

$x<-1$

№628 $\lim\limits_{n\to\infty}\left[\frac{x^{n+1}}{(n+1)!}+\frac{x^{n+2}}{(n+2)!}+\dots+\frac{x^{2n}}{(2n)!}\right]=☆$

$=\lim\limits_{n\to\infty} S_{2n}-S_n=0$

$S_{2n}=\sum\limits_0^{2n}\frac{x^k}{k!}\qquad e^x=\sum\limits_0^{\infty}\frac{x^k}{k!}$

$☆=\lim\limits_{n\to\infty}\underbrace{\frac{x^{n+1}}{(n+1)!}}_{\to 0}\left(1+\frac{x}{n+2}+\frac{x^2}{(n+2)(n+3)}+\dots+\frac{x^n}{(n+2)\dots(2n)}\right)=0$

№629 $\lim\limits_{n\to\infty}\left[(1+x)(1+x^2)(1+x^4)\dots(1+x^{2^n})\right]\cdot\frac{(1-x)}{1-x}\quad |x|<1 =$

$=\frac{1}{1-x}\lim\limits_{n\to\infty}\left(1-x^{4^n}\right)=\frac{1}{1-x}$

№630 $\lim\limits_{n\to\infty}\frac{\left(\cos\frac{x}{2}\cos\frac{x}{4}\dots\cos\frac{x}{2^n}\right)\cdot\sin\frac{x}{2^n}}{\sin\frac{x}{2^n}}=\lim\limits_{n\to\infty}\frac{\sin x}{2^n\sin\frac{x}{2^n}}=$

$\frac{1}{2}\sin\frac{x}{2^{n-1}}$

$\frac{1}{4}\sin\frac{x}{2^{n-2}}$

$=\frac{\sin x}{x}$

65 0 $\alpha = O(\beta)$ при $x \to x_0$

$$\lim_{x \to x_0} \frac{\alpha}{\beta} = A \ne 0, \ < \infty$$

$$A = 1 \Rightarrow \alpha \sim \beta$$

$\alpha = o(\beta)$ $x \to x_0$

$$\lim_{x \to x_0} \frac{\alpha}{\beta} = 0$$

Классификация точек разрыва функции

№676 $f(x) = \begin{cases} \frac{x^2-4}{x-2}, & x \ne 2 \\ A, & x = 2 \end{cases}$ $A = 4$

№677 $f(x) = \frac{1}{(1+x)^2}$ $x \ne -1$ т.р. II рода

$f(-1) =$

№687-700 №688 $y = \frac{x^2-1}{x^3-3x+2}$ $\frac{(x-1)(x+1)}{(x-1)^2(x+2)}$

$x=1$ – ~~устр. разр.~~ р II рода

$x=2$ – р II рода

№680 $y = \frac{\frac{1}{x} - \frac{1}{x+1}}{\frac{1}{x-1} - \frac{1}{x}}$

$x=0$ устр.

$x=1$ ~~устр.~~ о функц.

$x=-1$ т.р. II рода

$$\frac{1}{x(x+1)} : \frac{1}{x(x-1)} = \frac{x-1}{x+1}$$

№700 $y = \frac{1}{1-e^{\frac{1}{1-x}}}$

$x=0$ р. II рода

$x = 1-0 \to 0$, $1+0 \to 1$ $\}$ р. I рода (скачок)

№701 - 718

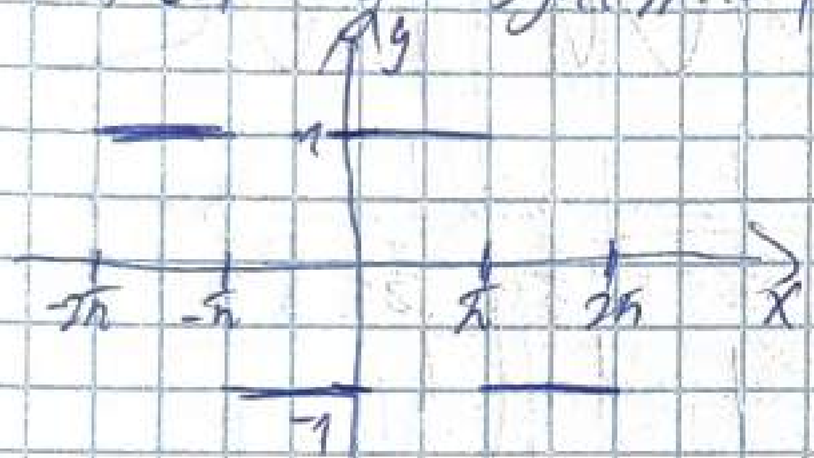
845 - 888

687 - 700

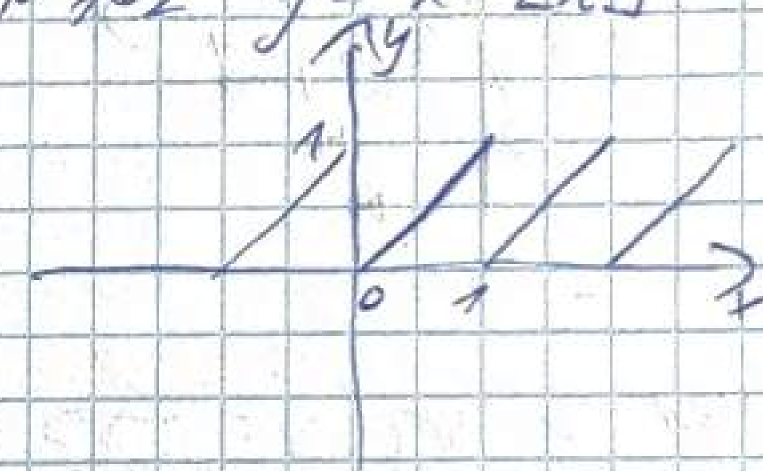
№883 $y = e^x + e^{e^x} + e^{e^{e^x}}$

$y' = e^x + e^x e^{e^x} + e^{e^{e^x}} e^{e^x} e^x$

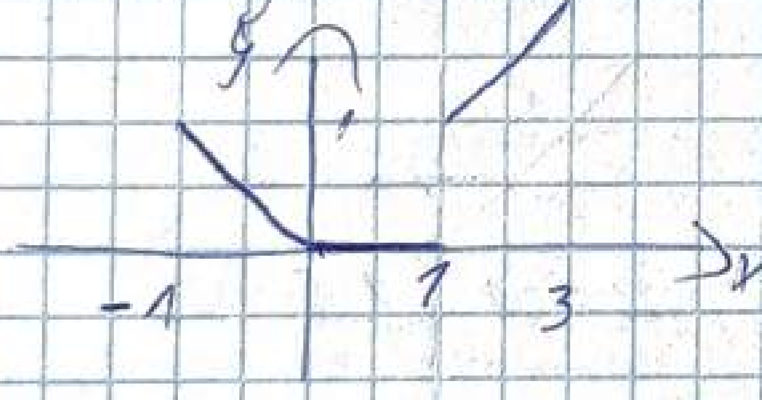
№701 $y = \operatorname{sgn}(\sin x)$



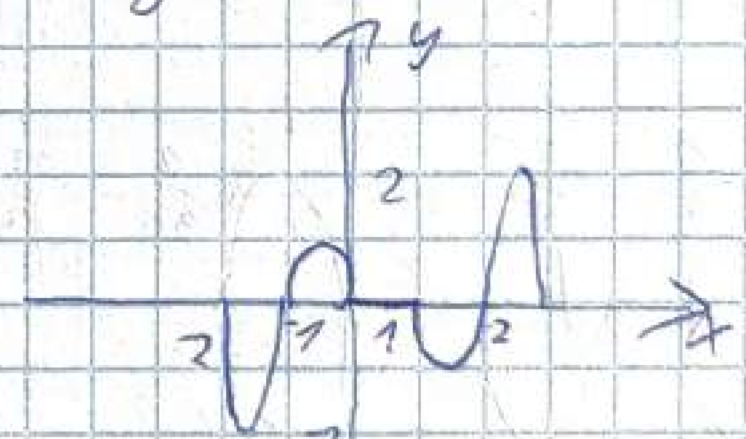
№702 $y = x - [x]$



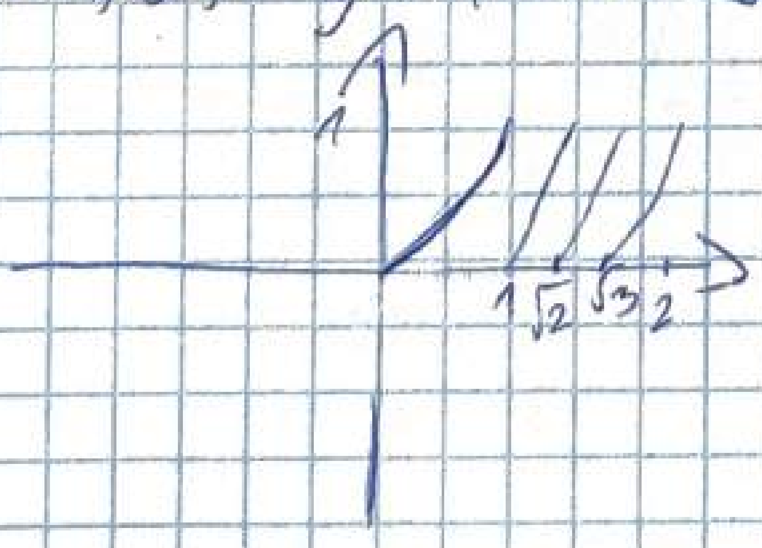
№703 $y = x[x]$



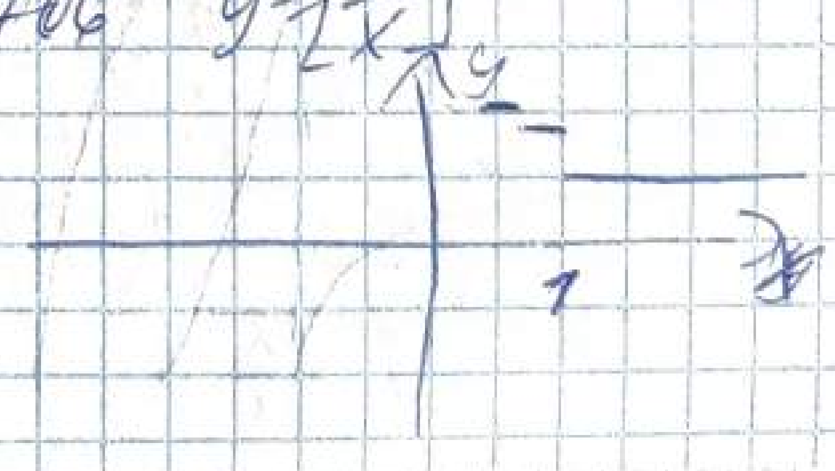
№704 $y = [x] \sin \pi x$



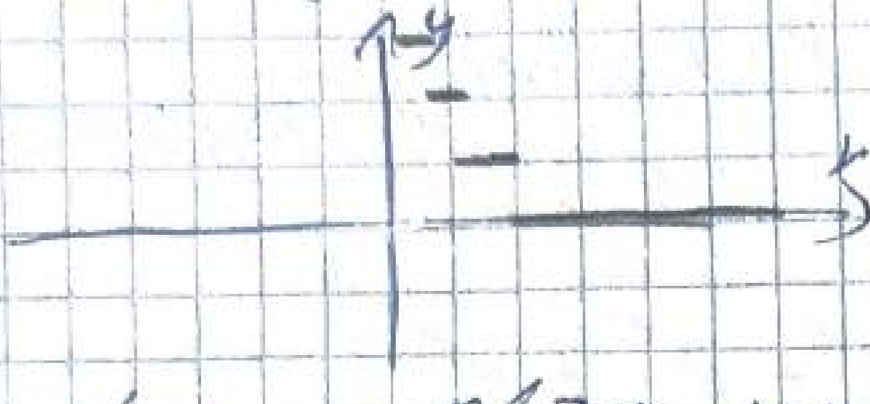
№705 $y = x^2 - [x^2]$



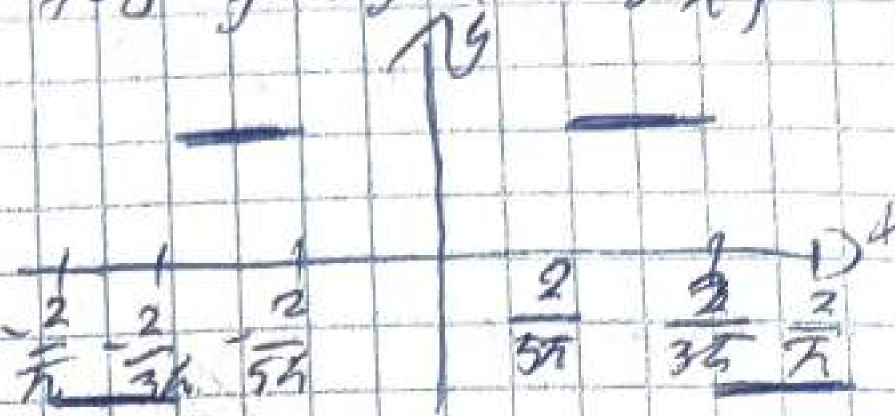
№706 $y = \left[\frac{1}{x}\right]$



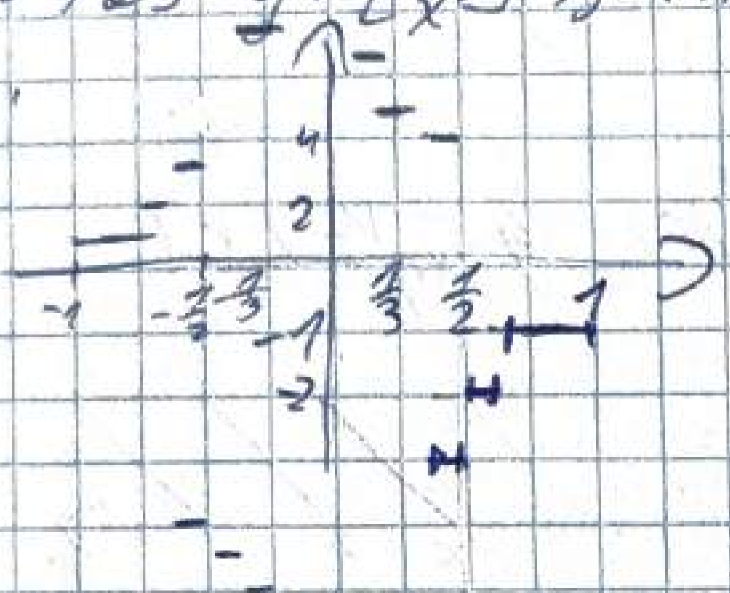
№707 $y = x\left[\frac{1}{x}\right]$?



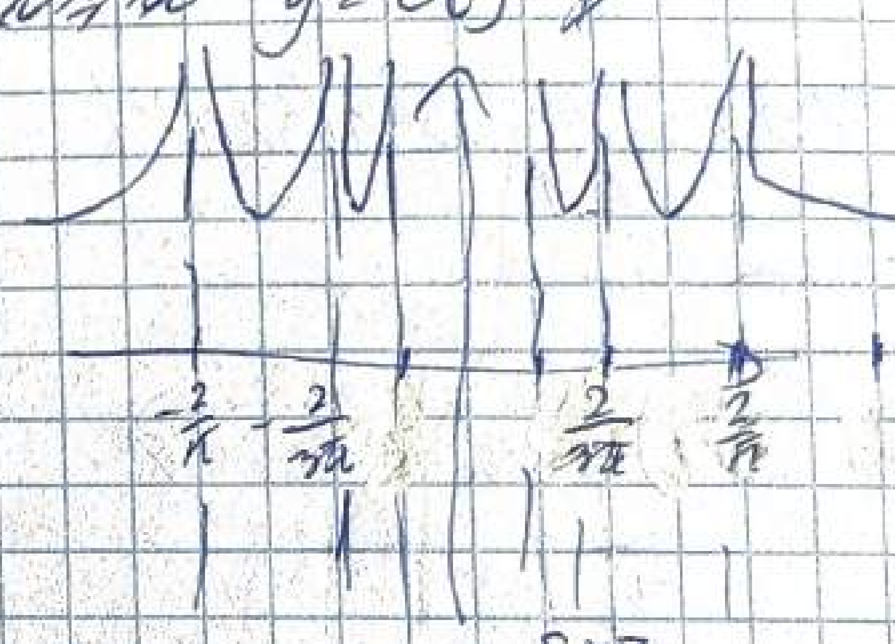
№708 $y = \operatorname{sgn}\left(\cos\frac{\pi}{x}\right)$



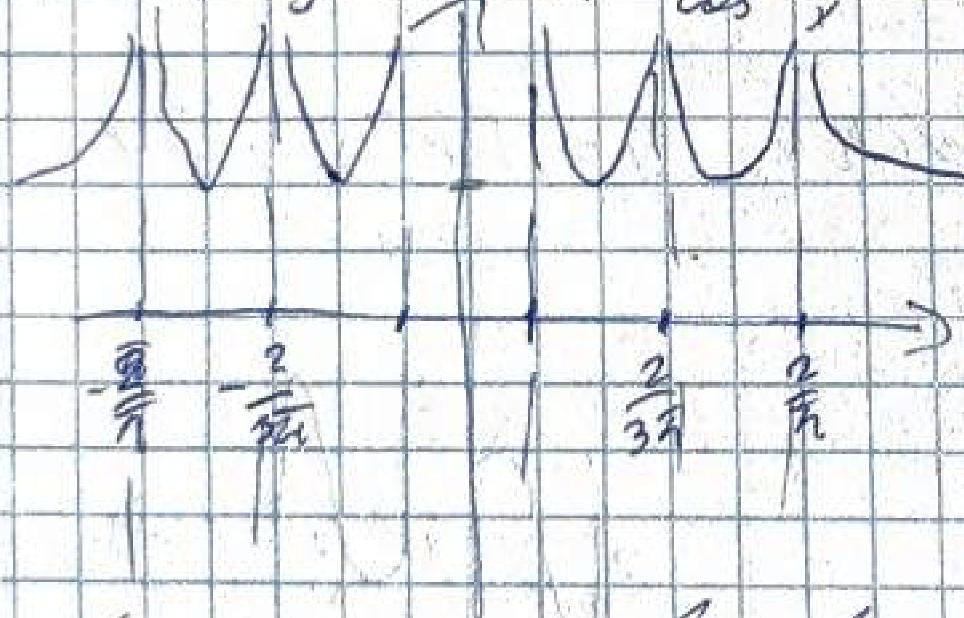
№709 $y = \left[\frac{1}{x^2}\right]\operatorname{sgn}\left(\sin\frac{\pi}{x}\right)$



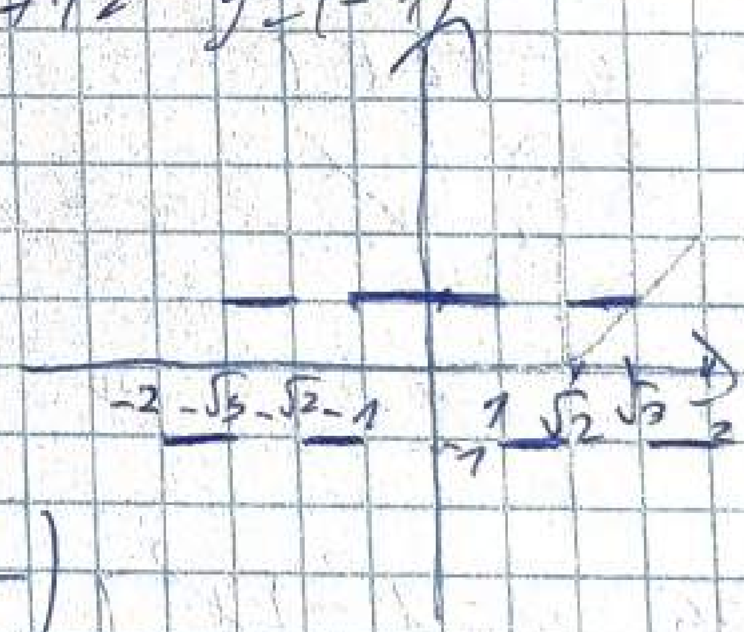
№710 $y = \operatorname{ctg}\frac{\pi}{x}$



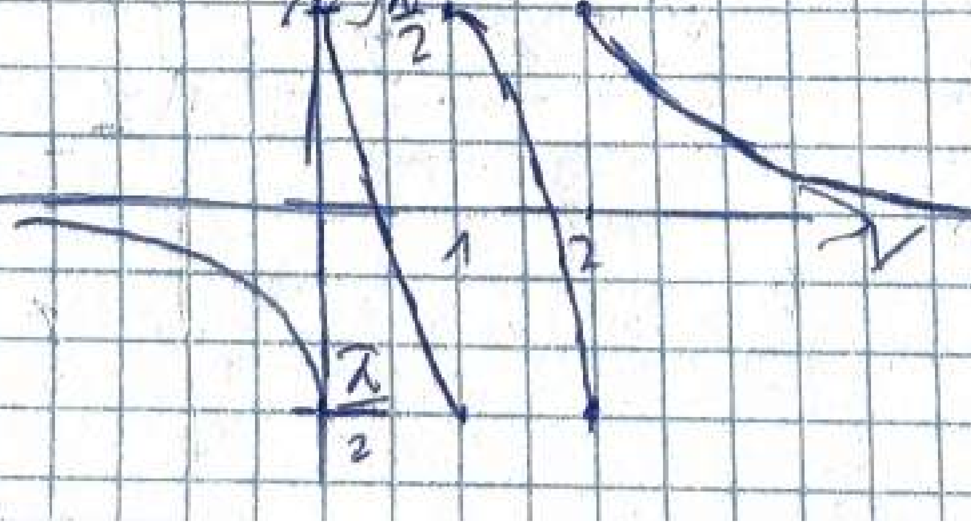
№711 $y = \sec^2\frac{\pi}{x} = \frac{1}{\cos^2\frac{\pi}{x}}$



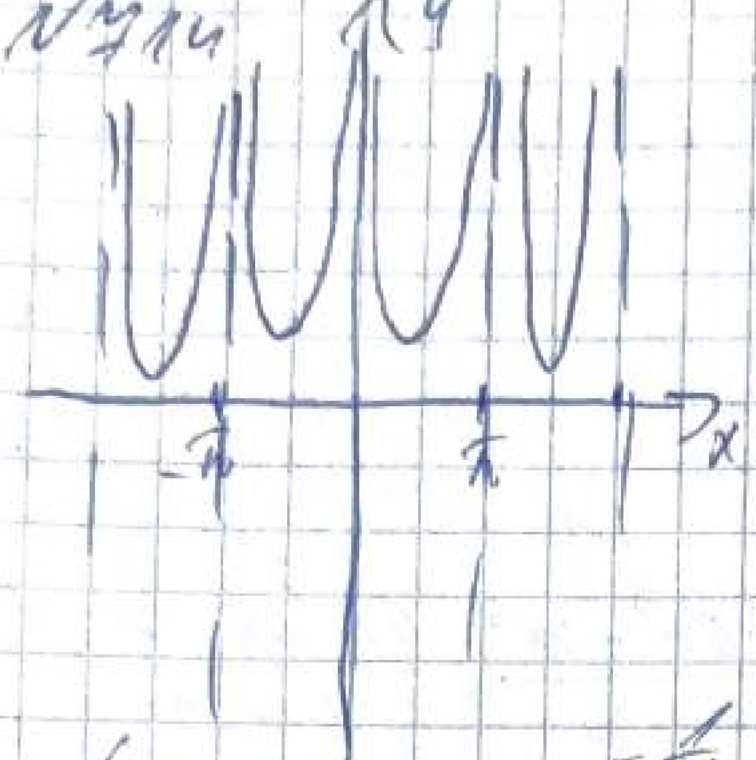
№712 $y = (-1)^{[x]}$



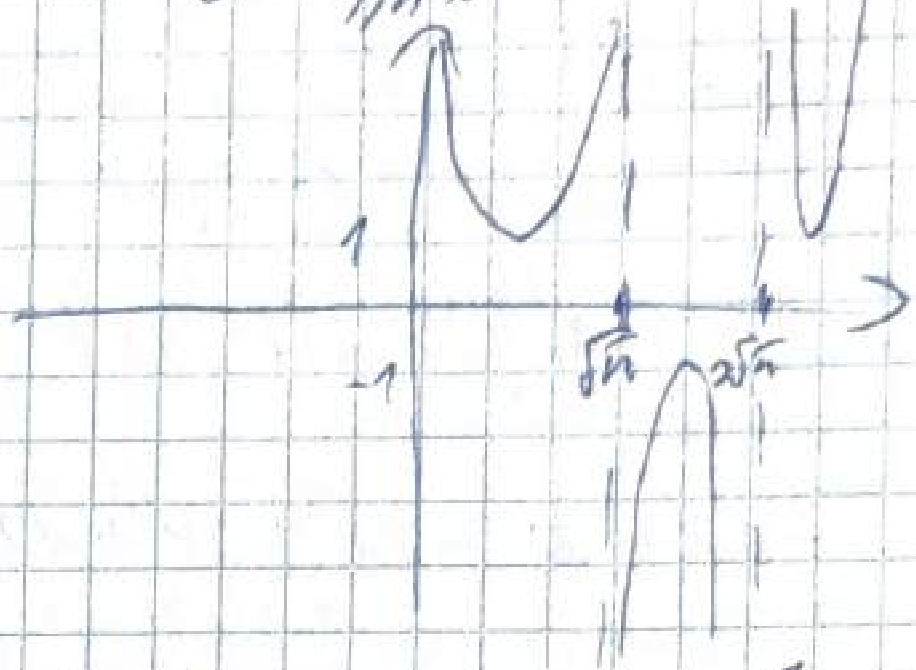
№713 $y = \operatorname{arcctg}\left(\frac{1}{x} + \frac{1}{x-1} + \frac{1}{x-2}\right)$



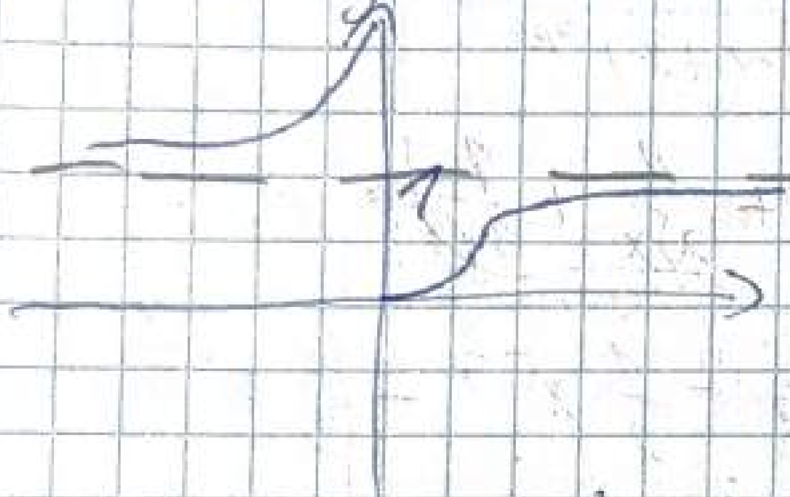
№714



№715 $y = \frac{1}{\sin x^2}$

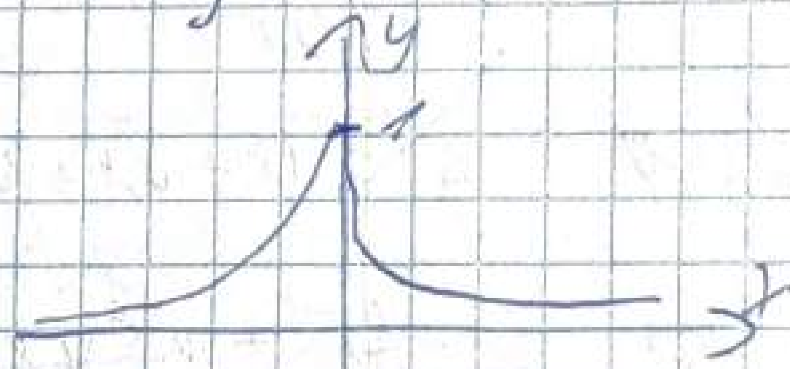


№717 $y = e^{-\frac{1}{x}}$



№718

$y = 1 - e^{-\frac{1}{x^2}}$



№845 $y' = \left(\frac{2x}{1-x^2}\right)' = \frac{2(1-x^2)+2x(2x)}{(1-x^2)^2} = \frac{2-2x^2+x^4}{(1-x^2)^2} = \frac{2(1+x^2)}{(1-x^2)^2}$

846) $y' = \frac{1+x-x^2}{1-x+x^2} = \frac{(1-2x)(1-x+x^2)-(-1+2x)(1+x-x^2)}{(1-x+x^2)^2} =$

$= \frac{\cancel{1-x+x^2-2x+2x^2-2x^3}\; 2(1-2x)}{(1-x+x^2)^2}$

847) $y' = \left(\frac{x}{(1-x)^2(1+x)^3}\right)' = \frac{(1-x)^2(1+x)^3 - x\left((1-x)(-2)(1+x)^3 + 3(1+x)^2(1-x)^2\right)}{(1-x)^4(1+x)^6} =$

$= \frac{(1-x)^2(1+x)^3+2x(1-x)-3(1+x)^2(1-x)^2}{(1-x)^4(1+x)^6} = \frac{1-x+4x^2}{(x+1)^4(1-x)^3}$

№848 $y' = \left(\frac{(2-x^2)(3-x^3)}{(1-x)^2}\right)' = \cancel{-2x(3-x^3)+(-3x^2)(2-x^2)}$

$= \frac{(1-x)^2((-2x)(3-x^3)-3x^2(2-x^2))+2(1-x)(2-x^2)(3-x^3)}{(1-x)^4} =$

$6-2x^3-3x^2+x^5$

$= \cancel{-2x(1+x^2)(3-x^3)-3x^2(1-}\ \frac{-2x(3-x^3)-3x^2(2-x^2)+2(2-x^2)(3-x^3)}{(1-x)^3}$

$= \frac{-6x+2x^3-6x^2+3x^4+12-4x^3-6x^2-2x^5}{(1-x)^3} = \frac{12-6x-12x^2-2x^3+3x^4-2x^5}{(1-x)^3}$

№849 $y' = \left(\frac{(1-x)^p}{(1+x)^p}\right)' = \frac{p(1-x)^{p-1}(1+x)^p - p(1+x)^{p-1}(1-x)^p}{(1+x)^{2p}}$

№851 $y' = (x+\sqrt{x}+\sqrt[3]{x})' = 1+\frac{1}{2\sqrt{x}}+\frac{1}{3\sqrt[3]{x^2}}$

№852 $y' = (x^{-1}+x^{-\frac{1}{2}}+x^{-\frac{1}{3}})' = -\frac{1}{x^2}-\frac{1}{2}x^{-\frac{3}{2}}-\frac{1}{3}x^{-\frac{4}{3}}$

№853 $y' = x^{\frac{2}{3}}-2x^{-\frac{1}{2}} = \frac{2}{3\sqrt[3]{x}}+\frac{1}{\sqrt{x^3}} = \frac{2}{3\sqrt[3]{x}}+\frac{1}{x\sqrt{x}}$

№854 $y' = (x(1+x^2)^{\frac{1}{2}})' = \cancel{\frac{1}{2}}\sqrt{(1+x^2)}+\frac{x^2}{\sqrt{1+x^2}} = \frac{1+2x^2}{\sqrt{1+x^2}}$

№855 $y' = ((1+x)(2+x^2)^{\frac{1}{2}}(3+x^3)^{\frac{1}{3}})' = (2+x^2)^{\frac{1}{2}}(3+x^3)^{\frac{1}{3}} +$

$+(1+x)\left(\frac{x(3+x^3)^{\frac{1}{3}}}{\sqrt{2+x^2}}+\frac{x(2+x^2)^{\frac{1}{2}}}{\sqrt[3]{(3+x^3)^2}}\right) = \cancel{(2+x^2)(1+x^3)^{\frac{2}{3}}} =$

$= \frac{(2+x^2)(1+x^3)^{\frac{3}{3}}+(1+x)x(3+x^3)^{\frac{3}{3}}+(1+x)\cancel{x}^2(2+x^2)}{\sqrt{2+x^2}\cdot\sqrt[3]{(3+x^3)^2}} =$

$$\frac{6+3x+5x^2+4x^3+2x^4+3x^5}{\sqrt{2+x^2}\cdot\sqrt[3]{(3+x^3)^2}}$$

№858

$$y'=\left(\sqrt[3]{\frac{1+x^3}{1-x^3}}\right)'=\frac{1}{3}\cdot\frac{1}{\sqrt[3]{\left(\frac{1+x^3}{1-x^3}\right)^2}}\cdot\frac{3x^2(1-x^3)+3x^2(1+x^3)}{(1-x^3)^2}=$$

$$=\frac{1}{\sqrt[3]{\left(\frac{1+x^3}{1-x^3}\right)^2}}\cdot\frac{3x^2-3x^5+3x^2+3x^5}{3(1-x^3)^2}=\left(\frac{1+x^3}{1-x^3}\right)^{-\frac{2}{3}}\cdot\frac{3x^2}{(1-x^3)^2}$$

№859

$$y'=\left(\frac{1}{\sqrt{1+x^2}\cdot(x+\sqrt{1+x^2})}\right)'=-\frac{2x+\frac{1}{2}\cdot 2x\cdot\frac{1}{\sqrt{1+x^2}}+\sqrt{1+x^2}}{(1+x^2+x\sqrt{1+x^2})^2}=$$

$$=-\frac{1}{(1+x^2)^{\frac{3}{2}}}$$

№860

$$y'=\left(\sqrt{x+\sqrt{x+\sqrt{x}}}\right)'=\frac{1}{2\sqrt{x+\sqrt{x+\sqrt{x}}}}\cdot\left(1+\frac{1}{2\sqrt{x+\sqrt{x}}}\cdot\left(1+\frac{1}{2\sqrt{x}}\right)\right)=$$

$$=\frac{4\sqrt{x}\sqrt{x+\sqrt{x}}+2\sqrt{x}+1}{8\sqrt{x}\cdot\sqrt{x+\sqrt{x}}\cdot\sqrt{x+\sqrt{x+\sqrt{x}}}}$$

№861

$$y'=\left(\sqrt[3]{1+\sqrt[3]{1+\sqrt[3]{x}}}\right)'=\frac{1}{3}\cdot\frac{1}{\sqrt[3]{(1+\sqrt[3]{1+\sqrt[3]{x}})^2}}\cdot$$

$$\cdot\left(\frac{1}{3}\cdot\frac{1}{\sqrt[3]{(1+\sqrt[3]{x})^2}}\cdot\frac{1}{3\sqrt[3]{x^2}}\right)=\frac{1}{27\sqrt[3]{(1+\sqrt[3]{1+\sqrt[3]{x}})^2}\cdot\sqrt[3]{(1+\sqrt[3]{x})^2}\cdot\sqrt[3]{x^2}}$$

№862

$$y'=(\cos 2x-2\sin x)'=-2\sin 2x-2\cos x=$$

$$=-2\cos x(1+\sin x)$$

№863 $y' = ((2-x^2)\cos x + 2x\sin x)' = -2x\cancel{\cos x} - \sin x(2-x^2) +$
$+ 2\sin x + \cancel{2x\cos x} = x^2\sin x$

№864 $y' = \sin(\cos^2 x)\cdot\cos(\sin^2 x) = \cos(\cos^2 x)\cdot$
$= \cos(\cos^2 x)\cdot 2\cos x\cdot(-\sin x)\cdot\cos(\sin^2 x) + 2\sin(\sin^2 x)\cdot\sin x\cdot\cos x\cdot\sin(\cos^2 x) =$
$= -\sin 2x(\cos(\cos^2 x)\cos(\sin^2 x) + \sin(\sin^2 x)\cos(\sin^2 x)) =$
$= -\sin 2x\cos(\cos 2x)$

№866 $y' = (\sin[\sin(\sin x)])' = \cos[\sin(\sin x)]\cdot$
$\cos(\sin x)\cdot\cos x$

№867 $y' = \left(\frac{\sin^2 x}{\sin x^2}\right)' = \frac{2\sin x\cos x\cdot\sin x^2 - \cos x^2\cdot 2x\cdot\sin^2 x}{\sin^2 x^2} =$

$= \frac{2\sin x(\cos x\sin x^2 - \cos x^2\cdot x\cdot\sin x)}{\sin^2 x^2}$

№868 $y' = \left(\frac{\cos x}{2\sin^2 x}\right)' = \frac{-2\sin^3 x - 4\sin x\cdot\cos^2 x}{4\sin^4 x} =$

$= -\frac{1+\cos^2 x}{2\sin^3 x}$

№870 $y' = \left(\frac{\sin x - x\cos x}{\cos x + x\sin x}\right)' = \cos x - \cos x + x\sin x$
$= \frac{(\cancel{\cos x} - \cancel{\cos x} + x\sin x)(\cos x + x\sin x) - (\cancel{\sin x} + \cancel{\sin x} + x\cos x)(\;)}{(\cos x + x\sin x)^2}$
$= \frac{(\cos x + x\sin x)(x\sin x - x\cos x)}{(\cos x + x\sin x)^2} = \frac{x(\sin x - \cos x)}{\cos x + x\sin x}$

№871 $y' = \left(tg\frac{x}{2} - ctg\frac{x}{2}\right)' = \frac{1}{2}\cdot\frac{1}{\cos^2\frac{x}{2}} + \frac{1}{2}\frac{1}{\sin^2\frac{x}{2}} = \frac{2}{\sin^2 x}$

№872 $y' = \left(tg\,x - \frac{1}{3}tg^3x + \frac{1}{5}tg^5x\right)' = \frac{1}{\cos^2 x} - \frac{tg^2x}{\cos^2 x} + \frac{tg^4x}{\cos^2 x} =$

№873 $y' = \left(4(ctg\,x)^{\frac{2}{3}} + (ctg\,x)^{\frac{8}{3}}\right)' = 4\cdot\frac{2}{3}\cdot ctg^{-\frac{1}{3}}x\cdot\left(-\frac{1}{\sin^2 x}\right) +$

$+ \frac{8}{3}ctg^{\frac{5}{3}}x\cdot\left(-\frac{1}{\sin^2 x}\right) = -\frac{8}{3}\cdot\frac{1}{\sin^2 x}\left(ctg^{-\frac{1}{3}}x + ctg^{\frac{5}{3}}x\right)$

№874 $y' = \overset{\sec^2\frac{x}{a}}{\frac{1}{\cos^2\left(\frac{x}{a}\right)}} + \overset{\mathrm{cosec}^2\frac{x}{a}}{\frac{1}{\sin^2\left(\frac{x}{a}\right)}} = \frac{x}{a}\sec^2\frac{x}{a}\cdot tg\frac{x}{a} + \frac{x}{a}\mathrm{cosec}^2\frac{x}{a}\cdot ctg\frac{x}{a}$

$= \frac{-16\cos\frac{2x}{a}}{a\sin^3\frac{2x}{a}}$

№875 $y' = \left(\sin\left[\cos^2\left(tg^3x\right)\right]\right)' = \cos\cos^2\left(tg^3x\right)\cdot$

$\cdot 2\cos\left(tg^3x\right)\cdot\left(-\sin tg^3x\right)\cdot\frac{1}{\cos^2x}\cdot 3tg^2x = \frac{-3tg^2x\sin\left(2tg^3x\right)}{\cos^2x}$

№876 $y' = \left(e^{-x^2}\right)' = -2xe^{-x^2}$

№877 $y' = \left(2^{tg\frac{1}{x}}\right)' = -\frac{1}{x^2}\cdot\frac{1}{\cos^2\frac{1}{x}}\cdot 2^{tg\frac{1}{x}}\ln 2$

878) $e^x\left(x^2-2x+2\right) = e^x\left(x^2-2x+2\right) + (2x-2)e^x = e^x x^2$

880) $e^x\left(1+ctg\frac{x}{2}\right) = e^x\left(1+ctg\frac{x}{2}\right) + \frac{1}{2}\cdot\left(-\frac{1}{\sin^2\frac{x}{2}}\right)e^x =$

$= e^x\left(1+ctg\frac{x}{2} - \frac{1}{2\sin^2\frac{x}{2}}\right)$

881) $\frac{\ln 3\sin x + \cos x}{3^x} = \frac{(\ln 3\cdot\cos x - \sin x)3^x - 3^x\ln 3(\ln^3\sin x + \cos x)}{3^{2x}} =$

$= \frac{\cancel{3^x\ln 3\cos x} - 3^x\sin x - 3^x\ln^2 3\sin x - \cancel{3^x\ln 3\cos x}}{3^{2x}} = -\frac{\sin x\left(1+\ln^2 3\right)}{3^x}$

№885 $\ln(\ln(\ln x))' = \frac{1}{\ln(\ln x)} \cdot \frac{1}{\ln x} \cdot \frac{1}{x}$

№888 $\ln(\ln^2(\ln^3 x))' = \frac{1}{\ln^2(\ln^3 x)} \cdot \frac{2\ln(\ln^3 x)}{\ln^3 x} \cdot \frac{3\ln^2 x}{x} =$

$= \frac{6\cancel{\ln(\ln^3 x)} \cdot \cancel{\ln^2 x}}{\ln^2(\ln^3 x) \cdot \ln^3 x \cdot x}$

$\lim\limits_{x\to 10} \frac{\lg x - 1}{x - 10} = \left| \begin{matrix} x - 10 = t \\ t \to 0 \end{matrix} \right| = \lim\limits_{t\to 0} \frac{\lg(10+t) - 1}{t} = \lim\limits_{t\to 0} \frac{\lg 10(1+\frac{t}{10}) - 1}{t} =$

$= \lim\limits_{t\to 0} \frac{1 + \lg(1+\frac{t}{10})}{t} = \frac{1}{10\lg 10}$

$\lim\limits_{x\to 0} (\cos 6x)^{\operatorname{ctg}^2 x} = 1^\infty = \lim\limits_{x\to 0} (1 + \cos 6x - 1)^{\frac{1}{\cos 6x - 1} \cdot \frac{(\cos 6x - 1)}{\operatorname{tg}^2 x}} =$

$= e^{\lim\limits_{x\to 0} \frac{1 - \cos 6x}{x^2}} = e^{-\frac{(6x)^2}{2x^2}} = e^{-18}$

$\lim\limits_{x\to 1} \cancel{\arcsin(\sqrt{x^2+x} + x)} \frac{\sqrt{x^2 - x + 1} - 1}{\operatorname{tg} \pi x} = \left| \begin{matrix} x - 1 = t \\ x = t + 1 \\ t \to 0 \end{matrix} \right| =$

$= \lim\limits_{t\to 0} \frac{\sqrt{t^2 + 2t + 1 - t - 1 + 1} - 1}{\operatorname{tg} \pi t} = \lim\limits_{t\to 0} \frac{t^2 + t}{\operatorname{tg} \pi t (\sqrt{t^2 + t + 1} + 1)} =$

$= \lim\limits_{t\to 0} \frac{\cancel{t}(t+1)}{\pi \cancel{t} \cdot 2} = \frac{1}{2\pi}$

$$\lim_{n\to\infty} n^{\frac{2}{n}} = e^{\lim\limits_{n\to\infty}\frac{2\ln n}{n}} = e^0 = 1$$

889 – 959 36, 35, 897,

№897

$$y = x\ln^2(x+\sqrt{1+x^2}) - 2\sqrt{1+x^2}\ln(x+\sqrt{1+x^2}) + 2x$$

$$y' = \ln^2(x+\sqrt{1+x^2}) + x\cdot 2\ln(x+\sqrt{1+x^2})\cdot\frac{1}{\sqrt{1+x^2}} -$$

$$-2\left(\frac{x}{\sqrt{1+x^2}}\cdot\ln(x+\sqrt{1+x^2}) - 2\cdot\sqrt{1+x^2}\cdot\frac{1}{\sqrt{1+x^2}}\right) + 2 =$$

$$= \ln^2(x+\sqrt{1+x^2})$$

№836

$$y = \frac{1}{4\sqrt2}\left(\ln(x^2+x\sqrt2+1) - \ln(x^2-x\sqrt2+1)\right) - \frac{1}{2\sqrt2}\operatorname{arctg}\frac{x\sqrt2}{x^2-1}$$

$$y' = \frac{1}{4\sqrt2}\left(\frac{2x+\sqrt2}{x^2+x\sqrt2+1} - \frac{2x-\sqrt2}{x^2-x\sqrt2+1}\right) - \frac{1}{2\sqrt2}\cdot\frac{1}{1+\frac{2x^2}{(1-x^2)^2}}\cdot$$

$$\cdot\frac{\sqrt2(x^2-1)-2\sqrt2x^2}{(x^2-1)^2} = \frac{1}{2}\cdot\frac{1-x^2}{x^4+1} + \frac{1}{2}\cdot\frac{x^2+1}{x^4+1} =$$

$$= \frac{1}{x^4+1}$$

$$\bullet \lim_{x\to 10} \frac{\lg x - 1}{x-10} = \left| \begin{array}{l} x-10=t \\ x=t+10 \\ t\to 0 \end{array} \right| = \lim_{t\to 0} \frac{\lg(10+t)-1}{t} = \lim_{t\to 0} \frac{\lg 10(1+\frac{t}{10})-1}{t}$$

$$= \frac{1}{10 \ln 10} =$$

$$\bullet \lim_{x\to\infty} \left(\sin\sqrt{1+x^2} - \sin\sqrt{x^2-1}\right) = \lim_{x\to\infty} 2\sin\frac{\sqrt{x^2+1}-\sqrt{x^2-1}}{2}\cdot\cos\frac{\sqrt{\ }+\sqrt{\ }}{2}$$

$$= \lim_{x\to\infty} 2\sin\underbrace{\frac{1}{\sqrt{x^2+1}+\sqrt{x^2-1}}}_{\to 0}\cdot\cos\frac{\sqrt{x^2+1}+\sqrt{x^2-1}}{2} = 0$$

$$\bullet \lim_{x\to 0} \frac{e^{x^2}-\cos x}{\sin^2 x} = \lim \frac{e^{x^2}-1}{x^2} - \frac{1-\cos x}{x^2} = \frac{x^2}{x^2} - \frac{x^2}{2x^2} = \frac{1}{2}$$

$$\bullet \lim_{x\to -\infty} \arcsin\left(\sqrt{x^2+x}+x\right) = \left| \begin{array}{l} -x=t \\ t\to+\infty \end{array} \right| = \lim_{t\to+\infty} \arcsin\left(\sqrt{t^2-t}-t\right) =$$

$$= \lim_{t\to+\infty} \arcsin\left(\frac{-t}{\sqrt{t^2-t}+t}\right) = \arcsin -\frac{1}{2} = -\frac{\pi}{2}$$

$$\bullet \lim_{x\to 0} \frac{(\sin 2x - 2\,\mathrm{tg}\,x)^2 + (1-\cos 2x)^3}{\mathrm{tg}^7 6x + \sin^6 x} = \cancel{\lim 4\sin^3 x}$$

$$= \lim_{x\to 0} \frac{4\sin^2 x\left(\cos x - \frac{1}{\cos x}\right)^2 + 8\sin^6 x}{(6x)^7 + x^6} = \lim_{x\to 0} \frac{4\sin^2 x\left(\frac{\cos^2 x-1}{\cos x}\right)^2 + 4\sin^4 x}{x^6(6^7 x+1)} =$$

$$= \lim \cancel{4\cdot\left(\frac{\sin^4 x}{\ }\right)} \lim_{x\to 0} \frac{4\left(\frac{\sin^4 x}{\cos^2 x} + 2\sin^4 x\right)}{x^4(6x+1)} = \frac{4\left(\frac{1}{\cos^2 x}+2\right)}{6x+1} =$$

$$= 12$$

• $y(x) = \dfrac{3^{\frac{1}{x}} + 2^{\frac{1}{x}}}{3^{\frac{1}{x}} - 2^{\frac{1}{x}}}$ т. разрыва

$$\lim_{x\to 0+0} \frac{3^{\frac{1}{x}}+2^{\frac{1}{x}}}{3^{\frac{1}{x}}-2^{\frac{1}{x}}} = \lim_{x\to 0+0} \frac{3^{\frac{1}{x}}(1+(\frac{2}{3})^{\frac{1}{x}})}{3^{\frac{1}{x}}(1-(\frac{2}{3})^{\frac{1}{x}})} = 1$$

$$\lim_{x\to 0-0} \frac{3^{\frac{1}{x}}+2^{\frac{1}{x}}}{3^{\frac{1}{x}}-2^{\frac{1}{x}}} = \left| \begin{array}{l} \frac{1}{x} = -t \\ t\to+\infty \end{array} \right| = \lim_{t\to+\infty} \frac{3^{-t}+2^{-t}}{3^{-t}+2^{-t}} \cdot 2^{-t} =$$

$$= \lim_{t\to+\infty} \frac{6^{-t}+1}{6^{-t}-1} = -1$$ скачок I рода

• $y(x) = \begin{cases} \dfrac{(1+x)^n - 1}{x}, & x \neq 0 \\ a, & x = 0 \end{cases}$ 889–939

$$\lim_{x\to 0} \frac{(1+x)^n - 1}{x} = \lim_{x\to 0} \frac{x\,n}{x} = n = a$$

№907 $y = \frac{1}{x}(\ln^3 x + 3\ln^2 x + 6\ln x + 6)$

$$y' = -\frac{1}{x^2}(\ln^3 x - 3\ln^2 x + 6\ln x + 6) + \frac{1}{x^2}(3\ln^2 x + 6\ln x + 6) =$$

$$= \frac{\ln^3 x}{x^2}$$

№906 $y = \ln \dfrac{b + a\cos x + \sqrt{b^2-a^2}\sin x}{a + b\cos x}$

$$y' = (\ln(b + a\cos x + \sqrt{b^2-a^2}\sin x) - \ln(a + b\cos x))' =$$

$$= \frac{-a\sin x + \sqrt{b^2-a^2}\cos x}{b + a\cos x + \sqrt{b^2-a^2}\sin x} + \frac{b\sin x}{a + b\cos x} =$$

$$= \frac{-a^2\sin x - ab\sin x\cos x + a\sqrt{b^2-a^2}\cos x + b\sqrt{b^2-a^2}\cos^2 x}{(a+b\cos x)(b + a\cos x + \sqrt{b^2-a^2}\sin x)} =$$

№ 919

$$y = x\arcsin\sqrt{\frac{x}{1+x}} + \operatorname{arctg}\sqrt{x} - \sqrt{x}$$

$$y' = \arcsin\sqrt{\frac{x}{1+x}} + \frac{x}{\sqrt{1-\sqrt{\frac{x}{x+1}}}} \cdot \frac{1}{2} \cdot \frac{1}{\sqrt{\frac{x}{1+x}}} \cdot \left((x+1)+x^2\right) +$$

$$+ \arcsin\sqrt{\frac{x}{1+x}} + \frac{1}{1+x} \cdot \frac{1}{2\sqrt{x}} - \frac{1}{2\sqrt{x}} =$$

$$= \frac{x(1+x)}{\sqrt{x}} + \arcsin\sqrt{\frac{x}{x+1}} + \frac{1}{2\sqrt{x}(1+x)}(1-x-x) =$$

$$= \arcsin\sqrt{\frac{x}{1+x}}$$

921

$$y = \frac{2}{\sqrt{a^2-b^2}}\operatorname{arctg}\left(\sqrt{\frac{a-b}{a+b}} \cdot \operatorname{tg}\frac{x}{2}\right)$$

$$y' = \frac{1}{a+b\cos x}$$

№961

$$\cancel{y = 1 + 1 + 1 + x^4}$$

$$y = x + x^{x} + x^{x^{x}} \qquad x > 0$$

$$y' = 1 + (x\ln x)' + x$$

№889 $y = \frac{1}{2}\ln(1+x) - \frac{1}{4}\ln(1+x^2) - \frac{1}{2(1+x)}$

$$y' = \frac{1}{2}\cdot\frac{1}{1+x} - \frac{\cancel{2}x}{\cancel{2}(1+x^2)} + \frac{(1+x)}{2(1+x)^2} = \cancel{\frac{x}{2(1+x^2)}} =$$

$$= \frac{1}{2(1+x)^2} - \frac{1}{2(1+x^2)} = \frac{1}{2(1+x^2)(x+1)^2}$$

№890 $y' = \left(\frac{1}{4}\ln\frac{x^2-1}{x^2+1}\right)' = \frac{1}{4}\cdot\frac{x^2+1}{x^2-1}\cdot\frac{2x(x^2+1)-2x(x^2-1)}{(x^2+1)^2} =$

$$= \frac{1}{\cancel{4}}\cdot\frac{x^2+1}{x^2-1}\cdot\frac{\cancel{2x^3}+2x-\cancel{2x^3}+2x}{(x^2+1)^2} = \frac{x(\cancel{x^2+1})}{(x^2-1)(x^2+1)^{\cancel{2}}} = \frac{x}{x^4-1}$$

№891 $y' = \left(\frac{1}{4(1+x^4)} + \frac{1}{4}\ln\frac{x^4}{1+x^4}\right)' = \frac{1}{\cancel{4}}\cdot(-1)\frac{\cancel{4}x^3}{(1+x^4)^2} +$

$$+ \frac{1}{\cancel{4}}\cdot\frac{1+x^4}{x^4}\cdot\frac{\cancel{4}x^3(1+x^4) - \cancel{4}x^3\cdot x^4}{(1+x^4)} =$$

$$= -\frac{x^3}{(1+x^4)^2} + \frac{1+x^4}{\cancel{4}x^4}\cdot\frac{\cancel{4}x^3 + \cancel{x^7} - x^7}{(1+x^4)^2} =$$

$$= -\frac{x^3}{(1+x^4)^2} + \frac{1}{x(1+x^4)} = \frac{1+x^4-x^4}{x(1+x^4)^2} = \frac{1}{x(1+x^4)^2}$$

№892 $y' = \left(\frac{1}{2\sqrt{6}}\ln\frac{x\sqrt{3}-\sqrt{2}}{x\sqrt{3}+\sqrt{2}}\right)' =$

$$=\frac{1}{2\sqrt{6}}\cdot\frac{x\sqrt{3}+\sqrt{2}}{x\sqrt{3}-\sqrt{2}}\cdot\frac{\sqrt{3}(x\sqrt{3}+\sqrt{2})-\sqrt{3}(x\sqrt{3}+\sqrt{2})}{(x\sqrt{3}-\sqrt{2})^2}=$$

$$=\frac{1}{\cancel{2\sqrt{6}}}\cdot\frac{x\sqrt{3}+\sqrt{2}}{x\sqrt{3}-\sqrt{2}}\cdot\frac{\cancel{x}+\sqrt{6}-\cancel{x}+\sqrt{6}}{3x^2-2}=\frac{1}{3x^2-2}$$

№894 $y'=(\sqrt{x+1}-\ln(1+\sqrt{x+1}))'=-\frac{1}{2\sqrt{x+1}}-$

$$-\frac{1}{1+\sqrt{x+1}}\cdot\left(-\frac{1}{2\sqrt{x+1}}\right)=-\frac{1}{2\sqrt{x+1}}+\frac{1}{2(\sqrt{x+1}+x+1)}=$$

$$=-\frac{1}{2\sqrt{x+1}}+\frac{1}{2\sqrt{x+1}(1+\sqrt{x+1})}=\frac{\sqrt{x+1}}{2\sqrt{x+1}(1+\sqrt{x+1})}=\frac{1}{2(1+\sqrt{x+1})}$$

№895 $y'=(\ln(x+\sqrt{x^2+1}))'=\frac{1}{x+\sqrt{x^2+1}}\cdot\left(1+\frac{2x}{2\sqrt{x^2+1}}\right)=$

$$=\frac{1}{x+\sqrt{x^2+1}}+\frac{x}{\sqrt{x^2+1}(x+\sqrt{x^2+1})}=\frac{x+\sqrt{x^2+1}}{\sqrt{x^2+1}(x+\sqrt{x^2+1})}=\frac{1}{\cancel{x+}\sqrt{x^2+1}}$$

№896 $y'=(x\ln(x+\sqrt{1+x^2})-\sqrt{1+x^2})'=$

$$=\ln(x+\sqrt{1+x^2})+\frac{x}{x+\sqrt{x^2+1}}\cdot\left(1+\frac{x}{\sqrt{1+x^2}}\right)-\frac{x}{\sqrt{1+x^2}}=$$

$$=\ln(x+\sqrt{1+x^2})+\frac{x}{x+\sqrt{x^2+1}}+\frac{x^2}{\sqrt{1+x^2}(x+\sqrt{1+x^2})}-\frac{x}{\sqrt{1+x^2}}=$$

$$=\ln(x+\sqrt{1+x^2})+\frac{x}{\sqrt{x^2+1}}-\frac{x}{\sqrt{1+x^2}}=\ln(x+\sqrt{1+x^2})$$

№900

$y'=\left(\frac{2+3x^2}{x^4}\cdot\sqrt{1-x^2}+3\ln\frac{1+\sqrt{1-x^2}}{x}\right)'=$

$$=\frac{6x\cdot 4x-4x^3(2+3x^2)}{x^8}\cdot\sqrt{1-x^2}+\frac{2+3x^2}{x^4}\cdot\left(-\frac{x}{\sqrt{1-x^2}}\right)+$$

$$+3\cdot\frac{x}{1+\sqrt{1-x^2}}\cdot\frac{-\frac{x}{2\sqrt{1-x^2}}\cdot x-1-\sqrt{1-x^2}}{x^2}=$$

$$=\frac{24x^2-8x^3-12x^5}{x^8}\cdot\sqrt{1-x^2}-\frac{2+3x^2}{x^3\sqrt{1-x^2}}+\frac{\frac{1}{2\sqrt{1-x^2}}-1-\sqrt{1-x^2}}{x(1+\sqrt{1-x^2})}=$$

$$=\frac{(24-8x-12x^3)\sqrt{1-x^2}}{x^6}-\frac{2+3x^2}{x^3\sqrt{1-x^2}}+\frac{-2-2x^2+1-2\sqrt{1-x^2}}{\sqrt{1-x^2}\,x(1+\sqrt{1-x^2})}=$$

$$y'=\frac{6x^5-4x^3(2+3x^2)}{x^8}\sqrt{1-x^2}+\frac{1}{2\sqrt{1-x}}\cdot\frac{2+3x^2}{x^4}+3\frac{x}{1+\sqrt{1-x^2}}\cdot$$

$$\cdot\left(1-\frac{x}{\sqrt{1-x^2}}\right)-\frac{3}{x}=\frac{6x^5-8x^3-12x^5}{x^8}\sqrt{1-x^2}-\frac{2+3x^2}{2x^3\sqrt{1-x^2}}+$$

$$+\frac{3x}{\sqrt{1-x^2}(1+\sqrt{1-x^2})}-\frac{3}{x}=+\frac{8+6x^2}{x^5}\cdot\sqrt{1-x^2}+\frac{2+3x^2}{2x^3\sqrt{1-x^2}}-$$

$$+\frac{3}{\sqrt{1-x^2}(1+\sqrt{1-x^2})}+\frac{3}{x}=\frac{2(8+6x^2)\sqrt{1-x^2}\cdot\sqrt{1-x^2}(1+\sqrt{1-x^2})+}{2x^5(1+\sqrt{1-x^2})\sqrt{1-x^2}}$$

$$+\frac{(2+3x^2)x^2(1+\sqrt{1-x^2})+6x^5+3x^4(1+\sqrt{1-x^2})\sqrt{1-x^2}}{}=$$

$$=\frac{8}{x^5\sqrt{1-x^2}}$$

№ 901 $y'=\left(\ln tg\frac{x}{2}\right)'=ctg\frac{x}{2}\cdot\frac{1}{2}\cdot\frac{x}{2}\cdot\frac{1}{\cos^2\frac{x}{2}}=$

$$=\frac{1}{2\sin\frac{x}{2}\cos\frac{x}{2}}=\frac{1}{\sin x}$$

№902 $y'=(\ln tg(\frac{x}{2}+\frac{\pi}{4}))' = ctg(\frac{x}{2}+\frac{\pi}{4})\cdot\frac{1}{\cos^2(\frac{x}{2}+\frac{\pi}{4})}\cdot\frac{1}{2} =$

$= \frac{1}{\sin(x+\frac{\pi}{2})} = \frac{1}{\cos x}$

№903 $y'=(\frac{1}{2}ctg^2 x+\ln\sin x)' = \frac{1}{\cancel{2}}\cdot\cancel{2}\cdot ctg x\cdot\frac{1}{-\sin^2 x} +$

$+\frac{\cos x}{\sin x} = -\frac{\cos x}{\sin x}\cdot\frac{1}{\sin^2 x}+\frac{\cos x}{\sin x} = \frac{\cos x\sin^2 x-\cos x}{\sin^3 x} = ctg^3 x$

№904 $y'=(\ln\sqrt{\frac{1-\sin x}{1+\sin x}})' = \sqrt{\frac{1+\sin x}{1-\sin x}}\cdot\frac{1}{2}\cdot\frac{-\cos x(1+\sin x)-\cos x(1-\sin x)}{(1+\sin x)^2} =$

$= \frac{\sqrt{1+\sin x}}{\sqrt{1-\sin x}}\cdot\frac{1}{\cancel{2}}\cdot\frac{-\cancel{2}\cos x}{(1+\sin x)^2} = \frac{-\cos x\sqrt{1+\sin x}}{\sqrt{1-\sin x}\,(1+\sin x)^2} =$

$= -\frac{1}{\cos x}$

№905 $y'=(-\frac{\cos x}{2\sin^2 x}+\ln\sqrt{\frac{1+\cos x}{\sin x}})' =$

$= -\frac{-\sin x(2\sin^2 x)+2\cdot 2\cdot\sin x\cos x(\cos x)}{4\sin^4 x} +$

$+\frac{\sqrt{\sin x}}{\sqrt{1+\cos x}}\cdot\frac{-\sin^2 x-\cos x(1+\cos x)}{\sin^2 x}\cdot\frac{1}{2}\sqrt{\frac{1+\cos x}{\sin x}}\cdot\sqrt{\frac{\sin x}{1+\cos x}}$

$= -\frac{\cancel{2}\sin^3 x+\cancel{2}\sin x\cos^2 x}{\cancel{2}\sin^4 x}+\frac{\sin x}{2(1+\cos x)}\cdot\left(-\frac{\sin^2 x+\cos^2 x+\cos x}{\sin^2 x}\right) =$

$= \frac{\sin^3 x+2\sin x\cos^2 x}{2\sin^4 x}-\frac{1}{2\sin x} = \frac{\cos^2 x}{\sin^3 x}$

№907 $y' = \left(\frac{1}{x}(\ln^3 x + 3\ln^2 x + 6\ln x + 6)\right)' =$

$= -\frac{1}{x^2}(\ln^3 x + 3\ln^2 x + 6\ln x + 6) + \frac{1}{x}\left(3\ln^2 x \cdot \frac{1}{x} + 6\ln x \cdot \frac{1}{x} +\right.$

$\left.+ 6 \cdot \frac{1}{x}\right) = -\frac{\ln^3 x}{x^2} - \cancel{\frac{3}{x^2}\ln^2 x} - \cancel{\frac{6}{x^2}\ln x} - \cancel{\frac{6}{x^2}} + \cancel{\frac{3}{x^2}\ln^2 x} +$

$+ \cancel{\frac{6}{x^2}\ln x} + \cancel{\frac{6}{x^2}} = -\frac{\ln^3 x}{x^2}$

№908 $y' = \left(\frac{1}{4x^4}\ln\frac{1}{x} - \frac{1}{16}x^4\right)' = \frac{1}{4}\cdot(-4)\cdot\frac{1}{x^5}\cdot\ln\frac{1}{x} +$

$+ \frac{x}{4x^4} - \frac{1}{16}\cdot 4x^3 = -\frac{1}{x^5}\ln\frac{1}{x} + \frac{1}{4x^5} - \frac{x^3}{4x^5} = -\frac{1}{x^5}\ln\frac{1}{x}$

№909 $y' = \left(\frac{3}{2}(1 - \sqrt[3]{1+x^2})^2 + 3\ln(1 + \sqrt[3]{1+x^2})\right)' =$

$= \frac{3}{2}\cdot 2x \cdot 2(1 - \sqrt[3]{1+x^2}) \cdot \frac{1}{3}\cdot(1+x^2)^{-\frac{2}{3}} + 3 \cdot \frac{1}{1+\sqrt[3]{1+x^2}} \cdot$

$\cdot \frac{1}{3}(1+x^2)^{-\frac{2}{3}} \cdot 2x = \frac{(1-\sqrt[3]{1+x^2})\, 2x}{\sqrt[3]{(1+x^2)^2}} + \frac{2x}{(1+\sqrt[3]{1+x^2})\cdot\sqrt[3]{(1+x^2)^2}} =$

$= \frac{2x}{1+\sqrt[3]{1+x^2}}$

№910 $y' = \left(\ln\left[\frac{1}{x} + \ln\left(\frac{1}{x} + \ln\frac{1}{x}\right)\right]\right)' = \frac{1}{\frac{1}{x} + \ln\left(\frac{1}{x} + \ln\frac{1}{x}\right)} \cdot$

$\cdot\left(-\frac{1}{x^2} + \frac{1}{\frac{1}{x} + \ln\frac{1}{x}}\cdot\left(-\frac{1}{x^2} + \frac{1}{\frac{1}{x}}\right)\right) = \frac{1}{\frac{1}{x} + \ln\left(\frac{1}{x} + \ln\frac{1}{x}\right)} \cdot$

$\cdot\left(-\frac{1}{x^2} + \frac{x-1}{x^2\left(\frac{1}{x} + \ln\frac{1}{x}\right)}\right) = \frac{1}{\frac{1}{x} + \ln\left(\frac{1}{x} + \ln\frac{1}{x}\right)} \cdot \left(\frac{-\frac{1}{x} - \ln\frac{1}{x} + x + 1}{x^2\left(\frac{1}{x} + \ln\frac{1}{x}\right)}\right) =$

№911 $y' = (x[\sin(\ln x) - \cos(\ln x)])' = \sin \ln x - \cos \ln x +$
$x(\cos(\ln x) \cdot \frac{1}{x} + \sin(\ln x) \cdot \frac{1}{x}) = 2\sin(\ln x)$

№912 $y' = (\ln \operatorname{ctg} \frac{x}{2}) - \cos x \cdot \ln \operatorname{tg} x)' = \operatorname{ctg} \frac{x}{2} \cdot \frac{1}{\cos^2 \frac{x}{2}} \cdot \frac{1}{2} -$
$-(-\sin x \cdot \ln \operatorname{tg} x + \cos x \cdot \operatorname{ctg} x \cdot \frac{1}{\cos^2 x}) = \frac{\operatorname{ctg} \frac{x}{2}}{2\cos^2 \frac{x}{2}} + \sin x \ln(\operatorname{tg} x) -$
$-\cos x \operatorname{ctg} x \cdot \frac{1}{\cos^2 x} = 2 \cancel{\frac{\sin \frac{x}{2}}{\cos \frac{x}{2}}} + \underline{\sin x \ln \operatorname{tg} x 2}$

№913 $y' = (\arcsin \frac{x}{2})' = \frac{1}{\sqrt{1 - \frac{x^2}{4}}} \cdot \frac{1}{2} = \frac{1}{\sqrt{4 - x^2}}$

№914 $y' = (\arccos \frac{1-x}{\sqrt{2}})' = -\frac{1}{\sqrt{1 - \frac{(1-x)^2}{2}}} \cdot -\frac{1}{\sqrt{2}} =$
$= \frac{1}{\sqrt{-x^2 + 2x + 1}}$

№915 $y' = (\operatorname{arctg} \frac{x^2}{a})' = \frac{1}{1 + \frac{x^4}{a^2}} \cdot \frac{2x}{a} = \frac{2x \quad a^2}{a(a^2 + x^4)} =$
$= \frac{2xa}{a^2 + x^4}$

№916 $y' = (\frac{1}{\sqrt{2}} \operatorname{arcctg} \frac{\sqrt{2}}{x})' = \frac{1}{\cancel{\sqrt{2}}} (-\frac{1}{1 + \frac{2}{x^2}}) \cdot \cancel{\sqrt{2}} \cdot (-\frac{1}{x^2}) =$
$= \frac{1}{x^2 + 2}$

№917 $y' = (\sqrt{x} - \operatorname{arctg} \sqrt{x})' = \frac{1}{2\sqrt{x}} - \frac{1}{(1+x)} \cdot \frac{1}{2\sqrt{x}} =$
$= \frac{1 + x - 1}{2\sqrt{x}(1+x)} = \frac{\sqrt{x}}{2(1+x)}$

№918 $y' = (x+\sqrt{1-x^2}\arccos x)' = 1 + \left(-\frac{x}{\sqrt{1-x^2}}\right)\cdot \arccos x +$

$+ \left(-\frac{1}{\sqrt{1+x^2}}\cdot\sqrt{1-x^2}\right) = 1 - \frac{x\arccos x}{\sqrt{1-x^2}} - \sqrt{\frac{1-x^2}{1+x^2}} =$

$= \cancel{\frac{\sqrt{1-x^4} - \sqrt{1+x^2}\cdot x\arccos x - (1-x^2)}{\sqrt{1-x^4}}} = -\frac{x\arccos x}{\sqrt{1-x^2}}$

№919

$\cancel{y' = \left(x\arcsin\sqrt{\frac{x}{1+x}} + \operatorname{arctg}\sqrt{x} - \sqrt{x}\right)'} =$

№920

$=$ $y' = \left(\arccos\frac{1}{x}\right)' = -\frac{1}{\sqrt{1+\frac{1}{x^2}}}\cdot\left(-\frac{1}{x^2}\right) =$

$= \frac{1}{x\sqrt{x^2+1}}$

№921 $y' = (\arcsin(\sin x))' = \frac{1}{\sqrt{1+\sin^2 x}}\cdot\cos x = \operatorname{sgn}(\cos x)$

№922 $y' = (\arccos(\cos^2 x))' = -\frac{1}{\sqrt{1-\cos^4 x}}\cdot 2\cos x(-\sin x) =$

$= \frac{\sin 2x}{\sqrt{1-\cos^4 x}} \overset{?}{=} \frac{2\operatorname{sgn}(\sin x)\cdot\cos x}{\sqrt{1+\cos^2 x}}$

№923 $y' = (\arcsin(\sin x - \cos x))' =$

$= \frac{1}{\sqrt{1-(\sin x-\cos x)^2}}\cdot(\cos x+\sin x) = \frac{\cos x+\sin x}{\sqrt{\sin 2x}}$

№924 $y' = (\arccos\sqrt{1-x^2})' = -\frac{1}{\sqrt{1-(1-x^2)}}\cdot\frac{(-2x)}{2\sqrt{1-x^2}} =$

$= \frac{\operatorname{sgn}(x)}{\sqrt{1-x^2}}$

№925 $y'=\left(\operatorname{arctg}\frac{1+x}{1-x}\right)'=\frac{1}{1+\frac{(1+x)^2}{(1-x)^2}}\cdot\frac{(1-x)+(1+x)}{(1-x)^2}=$

$=\frac{(1-x)^2}{(1+x)^2+(1-x)^2}\cdot\frac{2}{(1-x)^2}=\frac{1}{(1+x)^2}$

№926

$y'=\left(\operatorname{arctg}\left(\frac{\sin x+\cos x}{\sin x-\cos x}\right)\right)'=\frac{1}{1+\left(\frac{\sin x+\cos x}{\sin x-\cos x}\right)^2}\cdot$

$\times\left(\frac{(\cos x-\sin x)(\sin x-\cos x)-(\cos x+\sin x)^2}{(\sin x-\cos x)^2}\right)=$

$=\frac{1}{1+\left(\frac{\sin x+\cos x}{\sin x-\cos x}\right)^2}\cdot\frac{-2}{(\sin x-\cos x)^2}=\frac{-2}{2+\sin 2x}$

№928 $y'=\arcsin\frac{1-x^2}{1+x^2}=\frac{1}{\sqrt{1-\left(\frac{1-x^2}{1+x^2}\right)^2}}\cdot\frac{(-2x)(1+x^2)-2x(1-x^2)}{(1+x^2)^2}=$

$=\frac{-2x-2x^3-2x+2x^3}{(1+x^2)^2\sqrt{1-\left(\frac{1+x^2}{1+x^2}\right)^2}}=\frac{-4x}{(1+x^2)^2\sqrt{1-\left(\frac{1-x^2}{1+x^2}\right)^2}}=$

$=\frac{-4x}{(1+x^2)2|x|}=\frac{-2\operatorname{sgn}(x)}{1+x^2}$

№929 $y'=(\arccos^{-2}(x^2))'=-2\arccos^{-3}(x^2)\cdot$

$\cdot\left(-\frac{1}{\sqrt{1-x^4}}\right)\cdot 2x=\frac{4x}{\arccos^3(x^2)\sqrt{1-x^4}}$

№930

$y'=\left(\operatorname{arctg}x+\frac{1}{3}\operatorname{arctg}(x^3)\right)'=\frac{1}{1+x^2}+\frac{1}{3}\frac{3x^2}{1+x^6}=$

$$= \frac{1}{1+x^2} + \frac{x^2}{(1+x^2)(1-x^2+x^4)} = \frac{1+x^4}{1+x^6}$$

№931 $$y' = (\ln(1+\sin^2 x) - 2\sin x \cdot \operatorname{arctg}(\sin x))' =$$

$$= \frac{2\sin x\cos x}{1+\sin^2 x} - 2\left(\cos \operatorname{arctg}(\sin x) + \sin x \cdot \cos x \cdot \frac{1}{1+\sin^2 x}\right) = \frac{2\sin x\cos x - 2\sin x\cos x}{1+\sin^2 x} = -2\cos x \operatorname{arctg}(\sin x)$$

№932 $$y' = (\ln(\arccos\frac{1}{\sqrt{x}}))' = \frac{1}{\arccos\frac{1}{\sqrt{x}}} \cdot \left(-\frac{1}{\sqrt{1-\frac{1}{x}}}\right) \cdot$$

$$\cdot \left(-\frac{1}{2}\right) \cdot x^{-\frac{3}{2}} = \frac{1}{\arccos\frac{1}{\sqrt{x}}} \cdot \frac{1}{2\sqrt{x^3}\sqrt{1-\frac{1}{x}}} =$$

$$= \frac{1}{2\arccos\frac{1}{\sqrt{x}} \cdot x\sqrt{x-1}}$$

№935 $$y' = \left(\frac{1}{6}\ln\frac{(x+1)^2}{x^2-x+1} + \frac{1}{\sqrt{3}}\operatorname{arctg}\frac{2x-1}{\sqrt{3}}\right)' =$$

$$= \frac{1}{6} \cdot \frac{x^2-x+1}{(x+1)^2} \cdot \frac{2(1+x)^3-(2x-1)(1+x)^2}{(x^2-x+1)^2} +$$

$$+ \frac{1}{\sqrt{3}} \cdot \frac{1}{1+\frac{(2x-1)^2}{3}} \cdot \frac{2\sqrt{3} - \frac{1}{2\sqrt{3}}(2x+1)}{3} =$$

$$= \frac{3(x+1)^2}{2(x+1)^2(x^2-x+1)} + \frac{1 - \frac{1}{2\sqrt{x}}}{\sqrt{3}(x+(2x-1)^2)} = \frac{1}{1+x^3}$$

№936 $y' = \left(\frac{1}{4\sqrt{2}}\ln\frac{x^2+x\sqrt{2}+1}{x^2-x\sqrt{2}+1} - \frac{1}{2\sqrt{2}}\operatorname{arctg}\frac{x\sqrt{2}}{x^2-1}\right)' =$

$= \frac{1}{4\sqrt{2}}\left(\frac{2x+\sqrt{2}}{x^2+x\sqrt{2}+1} - \frac{2x-\sqrt{2}}{x^2-x\sqrt{2}+1}\right) - \frac{1}{2\sqrt{2}}\cdot\frac{1}{1+\frac{2x^2}{(x^2-1)^2}}\cdot$

$\cdot\frac{\sqrt{2}(x^2-1)-2\sqrt{2}x^2}{(x^2-1)^2} = \frac{\cancel{1}}{4\sqrt{2}}\left(\frac{1-x^2}{2(x^4+1)}\right) + \frac{1-x^2}{2(x^4+1)} = \frac{1}{x^4+1}$

№937 $y' = \left(x(\arcsin x)^2 + 2\sqrt{1-x^2}\arcsin x - 2x\right)' =$

$= (\arcsin x)^2 + x\cdot\arcsin x\cdot\frac{2}{\sqrt{1-x^2}} + 2\left(-\frac{x}{\sqrt{1-x^2}}\cdot\arcsin x +\right.$

$\left.+\frac{\sqrt{1-x^2}}{\sqrt{1-x^2}}\right) - 2 = \arcsin^2 x + \frac{2x\arcsin x}{\sqrt{1-x^2}} +$

$\mp\frac{2x\arcsin x}{\sqrt{1-x^2}} + 2 - 2 = \arcsin^2 x$

№938 $y' = \left(\frac{\arccos x}{x} + \frac{1}{2}\ln\frac{1-\sqrt{1-x^2}}{1+\sqrt{1-x^2}}\right)' =$

$= \cancel{-\frac{1}{x^2}\arccos x + \frac{1}{x\sqrt{1-x^2}}} + \frac{1}{2}\,\frac{+2x\cdot\frac{1}{\sqrt{1-x^2}}}{1-\sqrt{1-x^2}} -$

$-\frac{1}{2}\,\frac{-2x\cdot\frac{1}{\sqrt{1-x^2}}}{1+\sqrt{1-x^2}} \mp \frac{-\frac{x}{\sqrt{1-x^2}} + \arccos x}{x^2} =$

$= \frac{x}{\sqrt{1-x^2}(1-\sqrt{1-x^2})} + \frac{x}{\sqrt{1-x^2}(1+\sqrt{1-x^2})} + \frac{-\arccos x - \frac{x}{\sqrt{1-x^2}}}{x^2} =$

$= -\frac{\arccos x}{x^2}$

№939 $y' = \left(\operatorname{arctg}\sqrt{x^2-1} - \frac{\ln x}{\sqrt{x^2-1}}\right)' = \frac{1}{x+x^2-x} \cdot \frac{2x}{\sqrt{x^2-1}} \cdot \frac{1}{2} -$

$- \frac{\frac{1}{x}\sqrt{x^2-1} - \frac{1}{2}\frac{2x}{\sqrt{x^2-1}} \cdot \ln x}{x^2-1} = \cancel{\frac{1}{x\sqrt{x^2-1}}} - \cancel{\frac{1}{x\sqrt{x^2-1}}} +$

$+ \frac{x \ln x}{(x^2-1)^{\frac{3}{2}}} = \frac{x \ln x}{(x^2-1)^{\frac{3}{2}}}$

№940 $y' = \left(\frac{\arcsin x}{\sqrt{1-x^2}} + \frac{1}{2}\left(\ln(1-x) - \ln(1+x)\right)\right)' =$

$= \frac{\frac{1}{\sqrt{1-x^2}} \cdot \sqrt{1-x^2} - \frac{-2x}{2\sqrt{1-x^2}} \arcsin x}{1-x^2} + \frac{1}{2}\left(\frac{-1}{1-x} - \frac{1}{1+x}\right)$

$= \frac{1 + x\arcsin x}{1-x^2} + \frac{1}{2}\,\frac{-2}{1-x^2} = \frac{x \arcsin x}{(1-x^2)^{\frac{3}{2}}}$

№941 $y' = \left(\frac{1}{12}\left(\ln(x^4-x^2+1) - \ln(x^2+1)^2\right) - \frac{1}{2\sqrt{3}} \operatorname{arctg}\frac{\sqrt{3}}{2x^2-1}\right)' =$

$= \frac{1}{12}\left(\frac{4x^3-2x}{x^4-x^2+1} - \frac{2(x^2+1)\cdot 2x}{(x^2+1)^2}\right) + \frac{1}{2\sqrt{3}} \cdot \frac{1}{1+\frac{3}{(2x^2-1)^2}} \cdot \sqrt{3} \cdot$

$\cdot \frac{2\cdot 2x}{(2x^2-1)^2} = \frac{2x}{(2x^2-1)^2+3} \quad \frac{2x^3-x}{6(x^4-x^2+1)} - \frac{x}{3(x^2+1)^2} + \frac{2x}{(2x^2-1)^2+3} =$

$= \frac{1}{12}\left(\frac{4x^3-2x}{x^4-x^2-1} - \frac{4x}{x^2+1}\right) + \frac{2x}{(2x^2-1)^2+3} = \frac{x^3}{1+x^6}$

№942 $y' = \left(\frac{x^6}{1+x^{12}} - \operatorname{arctg} x^6\right)' = \frac{6x^5(1+x^{12}) - 12x^{11}\cdot x^6}{(1+x^{12})^2} -$

$- \frac{6x^5}{1+x^{12}} = \frac{6x^5 + 6x^{17} - 12x^{17}}{(1+x^{12})^2} + \frac{6x^5}{1+x^{12}} =$

$= \frac{6x^5 - 6x^{17} + 6x^5 + 6x^{17}}{(1+x^{12})^2} = \frac{12x^5}{(1+x^{12})^2}$

№943 $y = \ln \frac{1-\sqrt[3]{x}}{\sqrt{1+\sqrt[3]{x}+\sqrt[3]{x^2}}} + \sqrt{3}\operatorname{arctg}\frac{1+2\sqrt[3]{x}}{\sqrt{3}}$

$y' = -\frac{1}{1-\sqrt[3]{x}}\cdot\frac{1}{3}\cdot\frac{1}{\sqrt[3]{x^2}} - \frac{1}{\sqrt{1+\sqrt[3]{x}+\sqrt[3]{x^2}}}\cdot\frac{1}{2}\cdot\frac{1}{\sqrt{1+\sqrt[3]{x}+\sqrt[3]{x^2}}}\cdot$

$\cdot\left(\frac{1}{3}x^{-\frac{2}{3}} + \frac{2}{3}x^{-\frac{1}{3}}\right) + \cancel{\sqrt{3}}\,\frac{1}{\cancel{3}+\frac{(1+2\sqrt[3]{x})^2}{\cancel{3}}}\cdot\frac{1}{\cancel{\sqrt{3}}}\cdot 2\cdot\frac{1}{3}\cdot x^{-\frac{2}{3}} =$

$= -\frac{1}{3\sqrt[3]{x^2}(1-\sqrt[3]{x})} - \frac{1}{2(1+\sqrt[3]{x}+\sqrt[3]{x^2})}\left(\frac{1}{3\sqrt[3]{x^2}} + \frac{2}{3\sqrt[3]{x}}\right) +$

$+ \frac{\cancel{2}}{\cancel{2}\sqrt[3]{x^2}(\cancel{3}+(1+2\sqrt[3]{x})^2)} = -\frac{1}{(1-x)\sqrt[3]{x}}$

№944 $y' = \left(\operatorname{arctg}\frac{x}{1+\sqrt{1-x^2}}\right)' = \frac{1}{1+\frac{x^2}{(1+\sqrt{1-x^2})^2}}\cdot\cancel{(1+\sqrt{1-x^2})}$

$\cdot\frac{(1+\sqrt{1-x^2}) - x\left(-\frac{2x}{2\sqrt{1-x^2}}\right)}{(1+\sqrt{1-x^2})^2} = \frac{1+\sqrt{1-x^2}+\frac{x^2}{\sqrt{1-x^2}}}{(1+\sqrt{1-x^2})^2+x^2} =$

$$= \frac{\sqrt{1-x^2} + 1 - x^2 + x^2}{\sqrt{1-x^2}\,(1 + 1 - x^2 + x^2 + 2\sqrt{1-x^2})} = \frac{1}{2\sqrt{1-x^2}}$$

№ 946 $y = \frac{3-x}{2}\sqrt{1-2x+x^2} + 2\arcsin\frac{1+x}{\sqrt{2}}$

$$y' = \frac{1}{2}\left(-\sqrt{1-2x+x^2} + \frac{3-x}{2}\left(\frac{2x-2}{2\sqrt{1-2x+x^2}}\right)\right) +$$

$$+ 2 \cdot \frac{1}{\sqrt{1 - \frac{(1+x)^2}{2}}} \cdot \frac{1}{\sqrt{2}} = -\frac{1}{2}\sqrt{} + \frac{\overset{3x-3-x^2+x}{(3-x)(x-1)}}{4\sqrt{}} +$$

$$+ \frac{2}{\sqrt{2-1-2x-x^2}} = \frac{-x^2+4x-3}{4\sqrt{1-2x+x^2}} - \frac{1}{2\sqrt{1-2x+x^2}}^{12} + \frac{2^{\,14}}{\sqrt{1-2x-x^2}} =$$

$$= \frac{-x^2+4x-3-2+8}{4\sqrt{1-2x-x^2}} = \frac{x^2}{\sqrt{1-2x-x^2}}$$

№ 947 $y = \frac{1}{4}\ln(\sqrt[4]{1+x^4}+x) - \frac{1}{4}\ln(\sqrt[4]{1+x^4}-x) -$

$$-\frac{1}{2}\operatorname{arctg}\frac{\sqrt[4]{1+x^4}}{x}$$

$$y' = \frac{1}{4} \cdot \frac{1}{\sqrt[4]{1+x^4}+x} \cdot \left(\frac{4x^3}{4\sqrt[4]{(1+x^4)^3}} + 1\right) -$$

$$-\frac{1}{4} \cdot \frac{1}{\sqrt[4]{1+x^4}-x} \cdot \left(\frac{4x^3}{4\sqrt[4]{(1+x^4)^3}} - 1\right) - \frac{1}{2}\,\frac{1}{1+\frac{\sqrt[4]{1+x^4}}{x^2}} \cdot$$

$$\cdot \frac{x^3 \cdot x - \sqrt[4]{1+x^4}}{\sqrt[4]{(1+x^4)^3}} = \frac{1}{\sqrt[4]{1+x^4}}$$

№948 $y' = (\operatorname{arcctg}(tg^2x))' = \frac{1}{1+tg^4x}\cdot 2tgx\cdot\frac{1}{\cos^2x} =$

$= \frac{1}{1+\left(\frac{\sin x}{\cos x}\right)^4}\cdot\frac{2\sin x}{\cos^3x} = \frac{2\sin x\cos x}{\cos^4x+\sin^4x} = \frac{\sin 2x}{\cos^4x+\sin^4x}$

№949

$y = \sqrt{1-x^2}\cdot\ln\sqrt{\frac{1-x}{1+x}} + \frac12\ln\frac{1-\sqrt{1-x^2}}{1+\sqrt{1-x^2}} + \sqrt{1-x^2}+$

$+\arcsin x = \sqrt{1-x^2}\cdot\ln\sqrt{1-x} - \sqrt{1-x^2}\ln\sqrt{1+x} +$

$+\frac12\ln(1-\sqrt{1-x^2}) - \frac12\ln(1+\sqrt{1-x^2}) + \sqrt{1-x^2}+$

$\arcsin x$

$y' = +\frac{x}{\sqrt{1-x^2}}\left(-\frac{1}{\sqrt{1-x}}\cdot\frac12\frac{1}{\sqrt{1-x}} - \frac{1}{\sqrt{1+x}}\cdot\frac12\frac{1}{\sqrt{1+x}}\right)+$

$+\frac12\left(+\frac{1}{1-\sqrt{1-x^2}}\cdot\frac{1}{\cancel{2}}\cdot\frac{\cancel{2}x}{\sqrt{1-x^2}} + \frac{1}{1+\sqrt{1-x^2}}\cdot\frac{1}{\cancel{2}}\frac{\cancel{2}x}{\sqrt{1-x^2}}\right) -$

$-\frac{x}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1-x^2}} = \frac{x}{\sqrt{1-x^2}}\cdot\ln\sqrt{\frac{1-x}{1+x}} =$

$= \frac{x}{\sqrt{1-x^2}\cdot2}\left(+\frac{1}{1-x}+\frac{1}{1+x}\right) + \frac12\left(\frac{x}{\sqrt{1-x^2}(1-\sqrt{1-x^2})}^{(1+\sqrt{1-x^2})}\right. +$

$\left.+\frac{x}{\sqrt{1-x^2}(1+\sqrt{1-x^2})}^{(1-\sqrt{1-x^2})}\right) - \frac{x+1}{\sqrt{1-x^2}} - \frac{x}{\sqrt{1-x^2}}\ln\sqrt{\frac{1-x}{1+x}} =$

$= \frac{\cancel{2}x}{(1-x^2)\cancel{2}\sqrt{1-x^2}} + \frac{x}{\sqrt{1-x^2}\,x^2} - \frac{1+x}{\sqrt{1-x^2}} - \frac{x}{\sqrt{1-x^2}}\ln\sqrt{\frac{1-x}{1+x}} =$

$$= \frac{\sqrt{1-x^2}}{x} - \frac{x}{\sqrt{1-x^2}} \ln\sqrt{\frac{1-x}{1+x}}$$

№ 950 $y = x\,\mathrm{arctg}\,x - \frac{1}{2}\ln\sqrt{1+x^2} - \frac{1}{2}(\mathrm{arctg}\,x)^2$

$$y' = \mathrm{arctg}\,x + \frac{x}{1+x^2} - \frac{1}{2}\frac{2x}{1+x^2} - \frac{1}{2}\,\mathrm{arctg}\,x \cdot \frac{1}{1+x^2} =$$

$$= \frac{x^2}{1+x^2}\,\mathrm{arctg}\,x$$

№ 951 $y = \ln(e^x + \sqrt{1+e^{2x}})$

$$y' = \frac{1}{e^x + \sqrt{1+e^{2x}}} \cdot \left(e^x + \frac{1}{2}\frac{2e^{2x}}{\sqrt{1+e^{2x}}}\right) =$$

$$= \frac{1}{e^x+\sqrt{1+e^{2x}}} \cdot \left(e^x + \frac{e^{2x}}{\sqrt{1+e^{2x}}}\right) = \frac{1}{e^x+\sqrt{1+e^{2x}}} \cdot \frac{e^x\sqrt{1+e^{2x}} + e^{2x}}{\sqrt{1+e^{2x}}} =$$

$$= \frac{e^x\cancel{(\sqrt{1+e^{2x}} + e^x)}}{\cancel{(e^x+\sqrt{1+e^{2x}})}\sqrt{1+e^{2x}}} = \frac{e^x}{\sqrt{1+e^{2x}}}$$

№ 952 $y = \mathrm{arctg}(x + \sqrt{1+x^2})$

$$y' = \frac{1}{1+(x+\sqrt{1+x^2})^2} \cdot \left(1 + \frac{x}{\sqrt{1+x^2}}\right) = \frac{\sqrt{1+x^2}+x}{(1+(x+\sqrt{1+x^2})^2)\cdot\sqrt{1+x^2}} =$$

$$= \frac{\sqrt{1+x^2}+x}{(1+x^2+2x\sqrt{1+x^2}+1+x^2)\sqrt{1+x^2}} = \frac{\sqrt{1+x^2}+x}{2(1+x^2+x\sqrt{1+x^2})\sqrt{1+x^2}} =$$

$$= \frac{\cancel{\sqrt{1+x^2}+x}}{2\sqrt{1+x^2}\,\cancel{(\sqrt{1+x^2}+x)}\sqrt{1+x^2}} = \frac{1}{2(1+x^2)}$$

№954 $y=\frac{1}{4\sqrt{3}}(\ln(\sqrt{x^2+2}-x\sqrt{3})-\ln(\sqrt{x^2+2}+x\sqrt{3}))+$
$+\frac{1}{2}\operatorname{arctg}\frac{\sqrt{x^2+2}}{x}$

$$y'=\frac{1}{4\sqrt{3}}\left(\frac{\frac{x}{\sqrt{x^2+2}}-\sqrt{3}}{\sqrt{x^2+2}-x\sqrt{3}}-\frac{\frac{x}{\sqrt{x^2+2}}+\sqrt{3}}{\sqrt{x^2+2}+x\sqrt{3}}\right)+\frac{1}{2}\cdot\frac{1}{1+\frac{x^2+1}{x^2}}\cdot$$

$$\cdot\frac{\frac{x^2}{\sqrt{x^2+2}}-\sqrt{x^2+2}}{x^2}=\frac{1}{(2-x^2)\sqrt{x^2+2}}+\frac{-2}{8(2x^2+1)\sqrt{x^2+2}}=$$

$$=\frac{-1}{(2x^2+1)\sqrt{x^2+2}}+\frac{1}{(2-x^2)\sqrt{x^2+2}}=$$

№959 $y=\frac{1}{2\sqrt{2}}\operatorname{arctg}\frac{x\sqrt{2}}{\sqrt{1+x^4}}-\frac{1}{4\sqrt{2}}(\ln(\sqrt{1+x^4}-x\sqrt{2})-$
$-\ln(\sqrt{1+x^4}+x\sqrt{2}))$

$$y'=\frac{1}{2\sqrt{2}}\cdot\frac{1}{1+\frac{2x^2}{1+x^4}}\cdot\frac{\sqrt{2}\sqrt{1+x^4}-\frac{2x^3\cdot x\sqrt{2}}{\sqrt{1+x^4}}}{x^4+1}=\frac{1-x^4}{2(x^2+1)^2\sqrt{x^4+1}}+$$

$$+\frac{1}{4\sqrt{2}}\left(\frac{\frac{2x^3}{\sqrt{1+x^4}}-\sqrt{2}}{\sqrt{1+x^4}-x\sqrt{2}}-\frac{\frac{2x^3}{\sqrt{1+x^4}}+\sqrt{2}}{\sqrt{1+x^4}+x\sqrt{2}}\right)=-\frac{x^4-1}{2\sqrt{x^4+1}(x^2-1)^2}$$

$$=\frac{\sqrt{x^4+1}}{1-x^4}$$

№597 $y=\arccos(\sin x^2-\cos x^2)$

$$y'=-\frac{1}{\sqrt{1-(\sin x^2-\cos x^2)^2}}\cdot(\cos x^2\cdot 2x+\sin x^2\cdot 2x)=$$

$$= -\frac{2x(\sin x^2+\cos x^2)}{\sqrt{1-(\sin x^2-\cos x^2)^2}} = -\frac{2x(\sin^2 x^2+\cos x^2)}{\sqrt{\sin 2x^2}}$$

№ 958. $y = \arcsin(\sin x^2) + \arccos(\cos x^2)$

$$y' = \frac{\cos^2 x^2 \cdot 2x}{1-\sin^2 x^2} + \frac{\sin^2 x^2 \cdot 2x}{1-\cos^2 x^2} = 2x(\operatorname{sgn}(\cos x^2) + \operatorname{sgn}(\sin x^2))$$

~~№ 959~~

№ 961,1 $y' = \left(\frac{(\ln x)^x}{x^{\ln x}}\right)' = e^{x\ln x - \ln^2 x} =$

$$\ln y = \ln (\ln x)^x - \ln x^{\ln x} = x \ln\ln x - (\ln x)^2 =$$

$$= \frac{(\ln x)^x}{x^{\ln x}} \cdot \left(\ln\ln x + \frac{1}{\ln x} - \frac{2\ln x}{x}\right)$$

№ 965,2

$$y = \left[\frac{\arcsin(\sin^2 x)}{\arccos(\cos^2 x)}\right]^{\operatorname{arctg}^2 x}$$

$$\ln y = \operatorname{arctg}^2 x \overbrace{(\ln(\arcsin(\sin^2 x)) - \ln(\arccos(\cos^2 x))}^{?}$$

$$y' = \left(\frac{\arcsin(\sin^2 x)}{\arccos(\cos^2 x)}\right)^{\operatorname{arctg}^2 x} \left(\frac{2\operatorname{arctg} x}{(1+x^2)}(a) + \right.$$

$$\left. \operatorname{arctg}^2 x \cdot \frac{2\sin x\cos x}{\arcsin(\sin^2 x)} \cdot \frac{1}{\sqrt{1-\sin^4 x}} - \frac{-\sin 2x}{\arccos(\cos^2 x)\sqrt{1-\cos^4 x}}\right)$$

№971 $y=\frac{b}{a}x+\frac{2\sqrt{a^2-b^2}}{a}\operatorname{arctg}\left(\sqrt{\frac{a-b}{a+b}}\,\operatorname{th}\frac{x}{2}\right)$

$$y'=\frac{b}{a}+\frac{\cancel{2}\,\sqrt{a^2-b^2}}{a}\cdot\frac{1}{1+\frac{a-b}{a+b}\cdot\operatorname{th}^2\frac{x}{2}}\cdot\sqrt{\frac{a-b}{a+b}}\cdot\frac{1}{\operatorname{ch}^2\frac{x}{2}}\cdot\frac{1}{\cancel{2}}$$

$$=\frac{b}{a}+\frac{\cancel{2}(a-b)}{a}\cdot\frac{a+b}{(a+b)+(a-b)\operatorname{th}^2\frac{x}{2}}\cdot\frac{1}{\operatorname{ch}^2\frac{x}{2}}=$$

$$=\frac{b}{a}+\frac{a-b}{a}\cdot\frac{a+b}{(a+b)\operatorname{ch}^2\frac{x}{2}+(a-b)\operatorname{sh}^2\frac{x}{2}}=$$

$$=\frac{b}{a}+\frac{a^2-b^2}{a(a\operatorname{ch}x+b)}=\frac{ab\operatorname{ch}x+b^2+a^2-b^2}{a(a\operatorname{ch}x+b)}=$$

$$=\frac{b\operatorname{ch}x+a}{a\operatorname{ch}x+b}$$

№972 $y'=\left(\ln\left(\cos^2x+\sqrt{1+\cos^4x}\right)\right)'=$

$$=\left|u=\cos^2x\right|=\frac{\quad}{u+\sqrt{1+u^2}}$$

$$y'_u=\frac{1}{\sqrt{1+u^2}}=\frac{1}{\sqrt{1+\cos^4x}}$$

№ 324 $y=\frac{1}{2}\operatorname{arctg}(\sqrt[4]{1+x^4})+\frac{1}{4}\ln\frac{\sqrt[4]{1+x^4}+1}{\sqrt[4]{1+x^4}-1}$

$y'=\cancel{\frac{1}{2(1+\sqrt{1+x^4})}}\quad\left|\sqrt[4]{1+x^4}=t\right|=$

$=\cancel{\frac{1}{2(1+t^2)}}\quad y=\frac{1}{2}\operatorname{arctg}t+\frac{1}{4}(\ln(t+1)-\ln(t-1))$

$y'=\frac{1}{2(1+t^2)}+\frac{1}{4}\left(\frac{1}{t+1}-\frac{1}{t-1}\right)=$

$=\frac{1}{2(1+t^2)}+\frac{t-1-t-1}{4(t^2-1)}=\frac{1}{2(1+t^2)}-\frac{1}{2(t^2-1)}=$

$=\frac{1}{2}\left(\frac{t^2-1-1-t^2}{(t^4-1)}\right)=-\frac{1}{t^4-1}=-\cancel{\frac{\sqrt{1+x^4}}{x^4}}=-\frac{1\cdot t'_x}{x^4}$

$=-\frac{1}{x^4}\cdot\frac{\cancel{4}\cdot x^3}{\cancel{4}(1+x^4)^{\frac{3}{4}}}=\frac{1}{x(1+x^4)^{\frac{3}{4}}}$

№ ~~1033~~ 9

$x=\sqrt[3]{1-\sqrt{t}}\qquad y'_x=\frac{y'_t}{x'_t}$

№ 1040 $\begin{cases}x=\sin^2 t\\ y=\cos^2 t\end{cases}\qquad y'_x=\frac{-\sin 2x}{\sin 2x}=-1$

№ 1044

$\begin{cases}x=a(t-\sin t)\\ y=a(1-\cos t)\end{cases}\qquad y'_x=\frac{a\sin t}{a(1-\cos t)}=\frac{\cancel{2}\sin\frac{t}{2}\cos\frac{t}{2}}{\cancel{2}\sin^2\frac{t}{2}}=$

$=\operatorname{ctg}\frac{t}{2}$

№1048 $x^2+2xy-y^2=2x$ $\quad y'$ при $x=2, y=4$

$(2,0)$

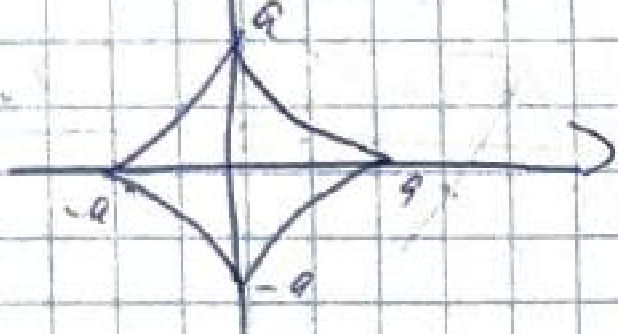
$\cancel{2}x+\cancel{2}y+\cancel{2}xy'-\cancel{2}yy'=\cancel{2}$

$y'(x-y)=1-x-y$

$y'=\frac{1-x-y}{x-y}$ $\quad y'(2,4)=\frac{-5}{-2}=\frac{5}{2}$

$y'(2,0)=-\frac{1}{2}$

№1052 $x^{\frac{2}{3}}+y^{\frac{2}{3}}=a^{\frac{2}{3}}$



астроида

$\frac{2}{3}x^{-\frac{1}{3}}+\frac{2}{3}y^{-\frac{1}{3}}y'=0$

$y'=-\frac{\sqrt[3]{y}}{\sqrt[3]{x}}$

№1053 $\operatorname{arctg}\frac{y}{x}=\ln\sqrt{x^2+y^2}$

$\begin{cases} y=r\sin\varphi \\ x=r\cos\varphi \end{cases}$ $\quad \varphi=\ln r$ $\quad r=e^{\varphi}$ $\quad \begin{cases} y=e^{\varphi}\sin\varphi \\ x=e^{\varphi}\cos\varphi \end{cases}$

$y'_x=\frac{e^{\varphi}\sin\varphi+e^{\varphi}\cos\varphi}{-e^{\varphi}\sin\varphi+e^{\varphi}\cos\varphi}=\frac{\sin\varphi+\cos\varphi}{\cos\varphi-\sin\varphi}$

$y'_x=\frac{x+y}{x-y}$

$$\frac{1}{1+\frac{y^2}{x^2}} \cdot \frac{y' \cdot x - y}{x^2} = \frac{1}{2} \frac{2x+2yy'}{x^2+y^2}$$

1039 - 1046

1048 - 1054

1055

$$xy' - y = x + yy'$$

$$y'(x-y) = x+y \qquad y' = \frac{x+y}{x-y}$$

№1054 $\quad y'_x$

a) $r = a\varphi$

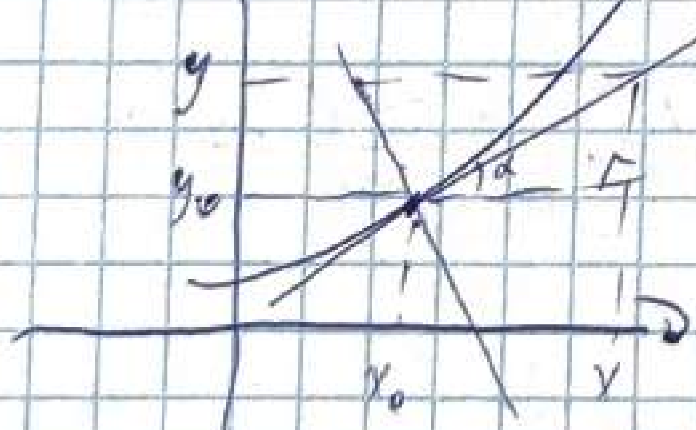
$$\begin{cases} x = a\varphi\cos\varphi \\ y = a\varphi\sin\varphi \end{cases} \quad y'_x = \frac{a\sin\varphi + a\varphi\cos\varphi}{a\cos\varphi - a\varphi\sin\varphi} = \frac{\sin\varphi + \varphi\cos\varphi}{\cos - \varphi\sin\varphi} =$$

$$= \frac{1+\varphi\,tg\,\varphi}{tg\,\varphi - \varphi}$$

$y = f(x) \quad , x = x_0$

$$y_{кас} = f(x_0) + f'(x_0)(x - x_0)$$

$$y_{нормаль} = f(x_0) - \frac{1}{f'(x_0)}(x - x_0)$$



$$f'(x_0) = \frac{y-y_0}{x-x_0} \Rightarrow y_к = f(x_0) + f'(x_0)(x-x_0)$$

$$\frac{y-y_0}{x_0-x_0} = tg\left(\pi - \left(\alpha + \frac{\pi}{2}\right)\right) = tg\left(\frac{\pi}{2} - \varphi\right) = \frac{1}{f'(x_0)}$$

1039 $\begin{cases} x = \sqrt[3]{1-\sqrt{t}} \\ y = \sqrt{1-\sqrt[3]{t}} \end{cases}$ $y'_x = \frac{y'_t}{x'_t}$

$x'_t = \frac{-\frac{1}{2}\frac{1}{\sqrt{t}}}{3(1-\sqrt{t})^{\frac{2}{3}}}$ $y'_t = \frac{-\frac{1}{3t^{\frac{2}{3}}}}{2\sqrt{1-\sqrt[3]{t}}}$

$y'_x = + \frac{\cancel{6}(1-\sqrt{t})^{\frac{2}{3}}\sqrt{t}}{\cancel{6}t^{\frac{2}{3}}\sqrt{1-\sqrt[3]{t}}} =$

№1041

$\begin{cases} x = a\cos t \\ y = b\sin t \end{cases}$ $x'_t = \cancel{\cos t} + a\sin t$

$y'_t = \cancel{\sin t} + b\cos t$

$y'_x = \frac{\sin t + b\cos t}{\cancel{\cos t} - a\sin t} = -\frac{b}{a}\,ctg\,t$

№1042

$\begin{cases} x = a\,ch\,t \\ y = b\,sh\,t \end{cases}$ $x'_t = a\,sh\,t$ $y'_t = b\,ch\,t$ $y'_x = \frac{b\,ch\,t}{a\,sh\,t} = \frac{b}{a}\,cth$

№1043

$\begin{cases} x = a\cos^3 t \\ y = a\sin^3 t \end{cases}$ $x'_t = -a\,3\cos^2 t\sin t$ $y'_t = a\,3\sin^2 t\cos t$ $y'_x = -\frac{\sin^{\cancel{2}}\cancel{\cos t}}{\cos^{\cancel{2}}\cancel{\sin t}} = -tg\,t$

№1044

$\begin{cases} x = e^{2t}\cos^2 t \\ y = e^{2t}\sin^2 t \end{cases}$ $x'_t = 2e^{2t}\cos^2 t - e^{2t}\,2\cos t\sin t$

$y'_t = 2e^{2t}\sin^2 t + e^{2t}\,2\sin t\cos t$

$$y'_x = \frac{e^{2t}\cos^2 t - e^{2t}\sin t\cos t}{e^{2t}\sin^2 t + e^{2t}\sin t\cos t} = \frac{\cos t(\cos t - \sin t)}{\sin t(\sin t + \cos t)} =$$

$$= \operatorname{ctg} t \, \frac{\cos t - \sin t}{\cos t + \sin t}$$

1046

$$\begin{cases} x = \arcsin \dfrac{t}{\sqrt{1+t^2}} \\ y = \arccos \dfrac{1}{\sqrt{1+t^2}} \end{cases} \quad x'_t = \frac{1}{\sqrt{1+\frac{t^2}{1+t^2}}} \cdot \left(\frac{\sqrt{1+t^2} - \frac{t^2}{\sqrt{1+t^2}}}{1+t^2} \right) =$$

$$= \frac{1+t^2-t^2}{\sqrt{1+t^2}} = \frac{1}{1+t^2}$$

$$(1+t^2)^{-\frac{1}{2}} = -\frac{1}{2}(1+t^2)^{-\frac{3}{2}} \cdot 2t$$

$$y'_t = -\frac{1}{\sqrt{1-\frac{1}{1+t^2}}} \cdot \cancel{\left(-\frac{1}{2}\frac{2t}{\sqrt{1+t^2}}\right) = \frac{t}{\sqrt{t^2}}}$$

$$\cdot \cancel{-2t} \quad \frac{\sqrt{1+t^2} - \frac{t}{\sqrt{1+t^2}}}{1+t^2} =$$

$$= -\frac{1}{\sqrt{1-\frac{1}{1+t^2}}} \cdot -\frac{t}{\sqrt{1+t^2}(1+t^2)} = \frac{t}{(1+t^2)|t|}$$

$$|t-1|^2$$

$$y'_x = \frac{t(1+t^2)}{|t|(1+t^2)} = \operatorname{sgn}(t)$$

1049 $y^2 = 2px$

$2yy' = 2p \qquad y' = \frac{p}{y}$

№1051 $\sqrt{x}+\sqrt{y}=\sqrt{a}$

$\frac{1}{2\sqrt{x}}+\frac{1y'}{2\sqrt{y}}=0 \qquad y'=-\frac{\sqrt{y}}{\sqrt{x}}$

№1050 $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

$\frac{2x}{a^2}+\frac{2yy'}{b^2}=0 \qquad \frac{yy'}{b^2}=-\frac{x}{a^2} \qquad y'=-\frac{b^2x}{a^2y}$

№1054

б) $r=a(1+\cos\varphi)$

$x=a(1+\cos\varphi)\cos\varphi=a\cos\varphi+a\cos^2\varphi$

$y=a(1+\cos\varphi)\sin\varphi=a\sin\varphi+a\cos\varphi\sin\varphi$

$y'_x=\frac{y'_\varphi}{x'_\varphi}=\frac{a\cos\varphi+a(\cos^2\varphi-\sin^2\varphi)}{\cancel{-a\sin\varphi-a2\cos\varphi\sin}\ -a\sin\varphi-a\sin2\varphi}=-\frac{\cos\varphi+\cos2\varphi}{\sin\varphi+\sin2\varphi}$

в) $r=ae^{m\varphi}$

$\begin{cases}x=ae^{m\varphi}\cos\varphi \\ y=ae^{m\varphi}\sin\varphi\end{cases} \qquad \begin{matrix}x'=a(me^{m\varphi}\cos\varphi-ae^{m\varphi}\sin\varphi) \\ y'=a(me^{m\varphi}\sin\varphi+ae^{m\varphi}\cos\varphi)\end{matrix}$

$y'_x=\frac{m\sin\varphi+\cos\varphi}{m\cos\varphi-\sin\varphi}$

№1055

$y=(x+1)\sqrt[3]{3-x}$

$y'=\sqrt[3]{3-x}+\frac{1(-1)(x+1)}{3(3-x)^{\frac{2}{3}}}=\sqrt[3]{3-x}-\frac{x+1}{3\sqrt[3]{(3-x)^2}}$

а) $A(-1,0)$

$f(-1)=0 \qquad f'(-1)=\sqrt[3]{4}$

$y_{кас}=\sqrt[3]{4}\,(x+1)$

$y_{норм}=-\dfrac{(x+1)}{\sqrt[3]{4}}=-\dfrac{\sqrt[3]{2}}{2}(x+1)$

б) $B(2,3)$

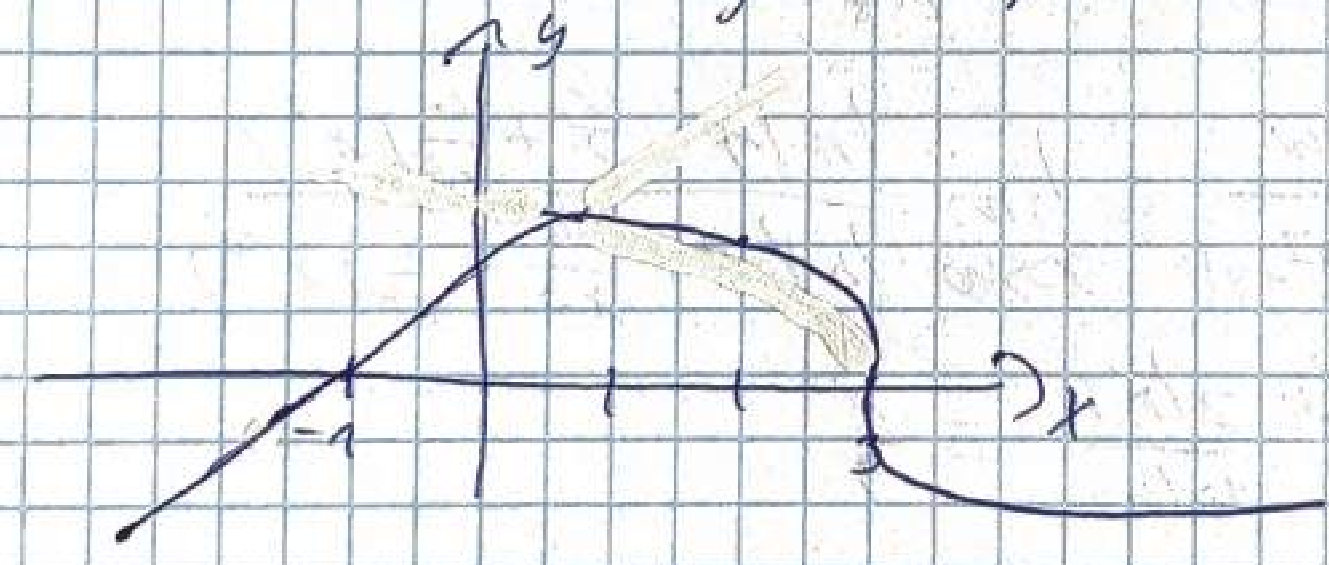
$y(2)=3 \qquad y'(2)=1-1=0$

$y_{кас}=3 \qquad$ ~~$y_{норм}$~~ $\quad x=2$ норм.

в) $C(3,0)$

$f(3)=0 \qquad f'(3)=\infty$

$x=3$ кас $\qquad y=0$ норм



• $n^{\frac{1}{4n}}=\sqrt[4n]{n}$

$n=\left(1+\left(\sqrt[4n]{n}-1\right)\right)^{4n}=1+4n\left(\sqrt[4n]{n}-1\right)+\dfrac{4n(n-1)}{2}\left(\sqrt[4n]{n}-1\right)+$

$+\ldots>\dfrac{2\cdot 4n(4n-1)}{8}\left(\sqrt[4n]{n}-1\right)^2$

$0\le\left(\sqrt[4n]{n}-1\right)<\sqrt{\dfrac{1}{4n-1}}\to 0$

$\sqrt[4n]{n}=1$

$$\bullet \lim_{x\to\frac{\pi}{2}} \left(x\,tg\,x - \frac{\pi}{2\cos x}\right) = \left| \begin{matrix} x-\frac{\pi}{2}=t \\ x=t+\frac{\pi}{2} \\ t\to 0 \end{matrix} \right| = \lim_{t\to 0} \left(\left(t+\frac{\pi}{2}\right)(-ctg\,t) + \right.$$

$$\left. + \frac{\pi}{2\sin t}\right) = \lim_{t\to 0} -\frac{1}{2}\,\frac{\cos t\,(2t+\pi)-\pi}{\sin t} =$$

$$= -\frac{1}{2}\lim_{t\to 0} \frac{\left(1-2\sin^2\frac{t}{2}\right)(2t+\pi)-\pi}{t} =$$

$$= -\frac{1}{2}\lim_{t\to 0} \frac{2t-4t\sin^2\frac{t}{2}-2\pi\sin^2\frac{t}{2}}{t} =$$

$$= -\frac{1}{2}\lim\left(2 - t^2\left(\frac{\sin\frac{t}{2}}{\frac{t}{2}}\right)^2 - \frac{2\pi}{t}\left(\frac{\sin\frac{t}{2}}{\frac{t}{2}}\right)^2\cdot\frac{t^2}{4}\right) =$$

$$= -\frac{1}{2}\lim_{t\to 0}\left(2-t^2-\frac{2\pi t^2}{2\cdot 4 t}\right) = -\frac{2}{2} = -1$$

$$\bullet \lim_{x\to+\infty} \frac{\ln(x^2-4x+4)}{\ln(x^{10}+5x^7+2)} = \lim_{x\to+\infty} \frac{\ln\left(\left(1-\frac{4}{x}+\frac{4}{x^2}\right)x^2\right)}{\ln\left(\left(1+\frac{5}{x^3}+\frac{2}{x^{10}}\right)x^{10}\right)} =$$

$$= \lim_{x\to+\infty} \frac{2\ln x}{10\ln x} = \frac{1}{5}$$

$$\bullet \lim_{x\to-\infty} \frac{\sqrt{x^2+4}-\sqrt[4]{4x^4+1}}{x} = \left| \begin{matrix} -x=t \\ t\to+\infty \end{matrix} \right| = \lim_{t\to\infty} \frac{\sqrt{t^2+4}-\sqrt[4]{4t^4+1}}{-t} =$$

$$= \lim_{t\to\infty} \frac{t\sqrt{1+\frac{4}{t^2}} - t\sqrt[4]{4+\frac{1}{t^4}}}{-t} = -(1-\sqrt{2}) = \sqrt{2}-1$$

$$\lim_{x\to 0}\left(\frac{2}{\sin 2x\,\sin^2 x}-\frac{1}{\sin^2 x}\right)=\lim_{x\to 0}\frac{2\sin x-\sin 2x}{\sin^2 x\,\sin 2x}=$$

$$=\lim_{x\to 0}\frac{2\cancel{\sin x}(\cancel{\sin x}\,1-\cos x)}{\sin^2 x\,\sin 2x}=\lim_{x\to 0}\frac{\cancel{x}(1-\cos x)}{x\cdot\cancel{x}\,x}=$$

$$=\lim_{x\to 0}\frac{2\sin^2\frac{x}{2}}{x^2}=\lim_{x\to 0}2\left(\frac{\sin\frac{x}{2}}{\frac{x}{2}}\right)^2\cdot\frac{x^2}{4x^2}=\frac{1}{2}$$